HR-S&I

Pan-European High-Resolution Snow & Ice Monitoring of the Copernicus Land Monitoring Service

Quality Assessment Report for Ice Products based on Sentinel-1 and Sentinel-2

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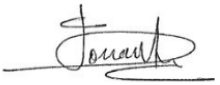
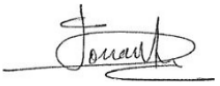
Checked by :	Joël Dorandeu	MAGELLIUM	01/11/2022	
Approved by :	Joël Dorandeu	MAGELLIUM	01/11/2022	

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1. Introduction

1.1. Scope and objectives

This document is the Quality Assessment Report (QAR) of the Pan-European **High-Resolution Snow & Ice Monitoring** (HR-S&I) of the Copernicus Land Monitoring Service (CLMS), dedicated to the scientific evaluation of the **ICE products**.

The Pan-European component of CLMS is coordinated by the European Environment Agency (EEA) and provides land cover, land use, ground-motion and biophysical products for the 32 EEA member countries and 6 cooperating countries in the Western Balkans, "EEA-38", and the United Kingdom jointly referred to as "EEA38+UK" (Figure 1). Part of the biophysical products, the HR-S&I data are describing the snow properties on land and ice occurrences on the hydrographic network.

Snow and ice cover monitoring are indeed critical parameters for a wide range of applications due to their strong influence on the water cycle and surface energy fluxes. The HR-S&I products came as a response to the specific needs of the European community. Mostly disseminated in near-real time (NRT), they are derived from high-resolution spatial observations, especially from synthetic aperture radar data from the Sentinel-1 constellation and from optical data from the Sentinel-2 constellation.

The purpose of this document is to provide an evaluation of the ice products derived from Sentinel-1 and Sentinel-2 observations.

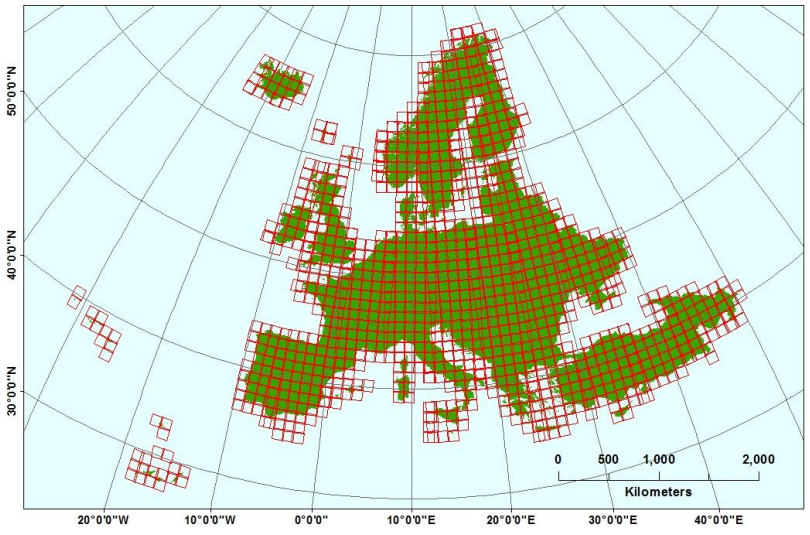


Figure 1. Sentinel-2 tiles over EEA38 +UK

1.2. Sentinel-1 and Sentinel-2 missions

Launched to cover the operational needs of the Copernicus Earth Observation programme, the Sentinel-1 (S1) and Sentinel-2 (S2) constellations are part of the European response to monitor the environment, mitigate the effects of climate change, and ensure civil security.

The S2 constellation is made up of two identical satellites (S2A and S2B, launched in June 2015 and March 2017 respectively). Each satellite carries a camera that images Earth's surface in 13 spectral bands at high spatial resolution (10 m for visible and near-infrared bands, 20 m for red edge and 60 m for shortwave infrared bands). They orbit the Earth at an altitude of 786 km but 180° apart, a configuration which optimises coverage and revisit times: every 5 days at the equator under cloud-free conditions which results in 2-3 days at European latitudes. Due to the overlap between S2A and S2B swaths from adjacent orbits, this revisit frequency increases under different viewing conditions.

The S2 Level 1C product (monodate orthorectified image expressed in top-of-atmosphere reflectance) is used to derive HR-S&I products. The L1C product is disseminated in granules of a fixed size: 110 km by 110 km tiles in UTM/WGS84 map projection (see Figure 1 for the tiles over EEA38+UK).

The S1 mission is a polar-orbiting satellite constellation for the continuation of operational Synthetic Aperture Radar (SAR) application. The S1 mission is made up of two identical satellites. The first of the dual S1 satellites, S1A, was launched in April 2014, and its identical twin, S1B, was launched in April 2016. The C-band SAR operates at single (HH or VV) or dual polarisation (HH+HV or VV+VH) and enables continuous acquisitions at all weather conditions and independent of day and night. The S1 Interferometric Wide (IW) swath mode is the primary acquisition mode over land area with a resolution of 5 m in range and 20 m in azimuth and a swath width of about 250 km. They orbit the Earth at an altitude of 693 km, 180° apart, operating in a repeat pass interval of 6 days. Since the orbit track spacing varies with latitude, the revisit rate is significantly greater at higher latitudes than at the equator (~2 days for Europe).

The S1 Level 1 Ground Range Detected (GRD) High Resolution (HR) data acquired in IW mode is used to generate HR-S&I products. It consists of focused SAR data that has been detected, multi-looked and projected to ground range using the Earth ellipsoid model WGS84. Both ascending (afternoon acquisitions) and descending (morning acquisitions) tracks are used.

Both, S1 and S2 constellations, provide valuable information for monitoring snow and ice over continental surfaces at high spatial resolutions. The main challenge when using the S2 optical data is to distinguish snow/ice from cloud cover. Once clouds are masked out, it is possible to detect the presence of snow over lands and ice over inland waters. The S1 radar technology allows observing the snow conditions (wet or dry snow) giving the possibility to track snow melting across Europe. The synergy between radar and optical data not only enables detailed information with high spatial resolution but more frequent and more complete products.

1.3. HR-S&I ice products summary

Ice occurrences on the hydrographic network are described by two types of products.

- The **R**iver and **L**ake **I**ce **E**xtent (RLIE) product provides pixel-based information about ice presence on rivers and lakes. There are several different RLIE products available depending on their data source (either S1, S2 or a combination of both types of observations).
- The **A**ggregated **R**iver and **L**ake **I**ce **E**xtent (ARLIE) product provides the percent coverage of snow-covered or snow-free ice on lakes and on 10 km river sections. This spatially aggregated product is also available in two versions depending on their data source.

The production chain of HR-S&I ice products is presented in Figure 2. It highlights the dependency between the products as well as the satellite data used to derive them.
NB: The aggregated ARLIE S2 product based on S2 data only will eventually not be maintained and replaced by the more comprehensive ARLIE S1+S2 product which relies on both types of observations.

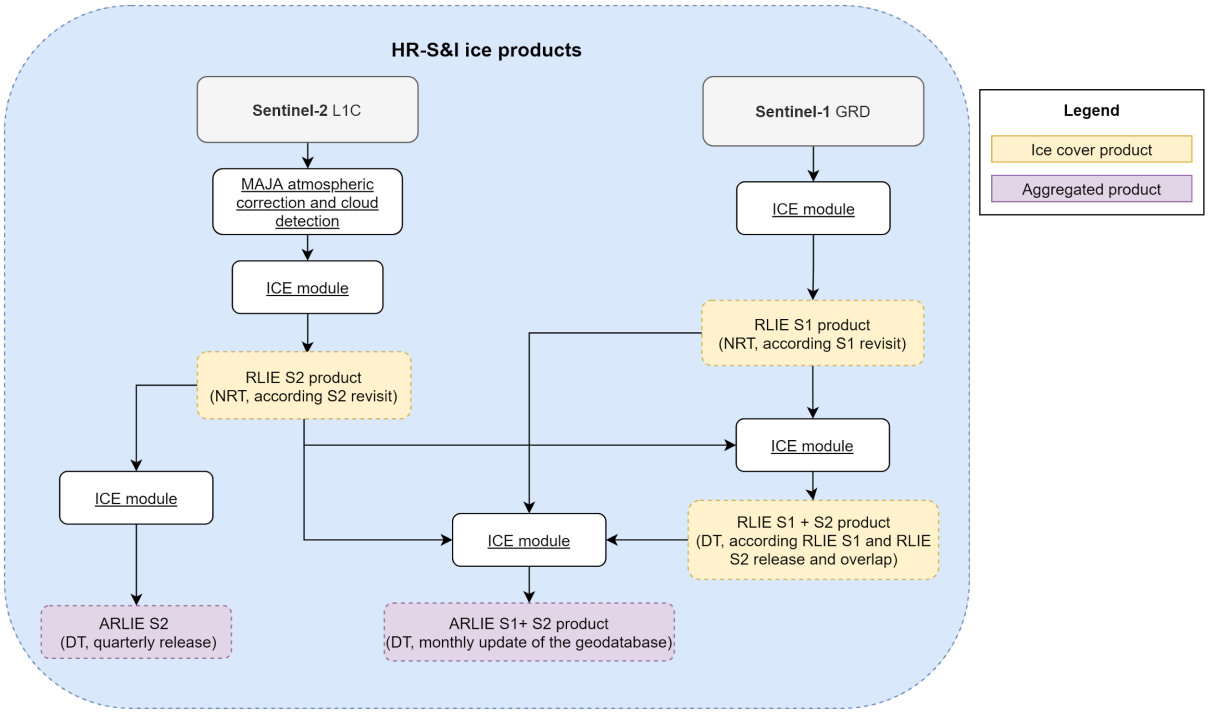


Figure 2. HR-S&I ice production workflow (NRT: near-real time, DT: delayed-time)

This document focuses on the evaluation of the **RLIE products** directly derived from S2 and S1 observations (**RLIE S2 and RLIE S1**). As the RLIE S1+S2 product and the ARLIE product are defined as a straightforward combination/aggregation of individual RLIE products, their quality directly depends on RLIE product quality. For information, the assessment of RLIE S2 and RLIE S1 have been performed independently, as RLIE S2 products were implemented first.

Before reading this document, users are invited to consult the documents [AD3, AD4] which respectively provide a description of the algorithms used for the RLIE S2 and RLIE S1 product generation, and the user manual [AD2] which depicts the content and format of the products. For the evaluation of other HR-S&I snow products, users should refer to the two quality assessment reports [AD1, AD5].

1.4. Document structure

In addition to this introduction, the document is organised as follows:

- Section 2 provides a detailed description of data used to evaluate the ICE algorithms together with a presentation of the evaluation metrics.
- Section 3 presents the quality assessment results for the ICE algorithms.
- Section 4 summarises the main conclusions of the quality assessment.

1.5. Related documents

1.5.1. Applicable documents

The following table lists the documents with a direct bearing on the content of this document.

Id.	Ref.	Description
AD1	COSIMS-DT-070-MAG_QAR_SNOW	Quality Assessment Report for snow products based on Sentinel-2
AD2	COSIMS-DT-061-MAG_PUM_ICE	Product User Manual for ice products
AD3	COSIMS-DT-063-MAG_ATBD_ICE	Algorithm Theoretical Basis Document for ice products based on Sentinel-2
AD4	COSIMS-DT-126-MAG_ATBD_ICE_S1	Algorithm Theoretical Basis Document for ice products based on Sentinel-1 and Sentinel-2
AD5	COSIMS-DT-070-MAG_QAR_SNOW_S1	Quality Assessment Report for snow products based on Sentinel-1 and Sentinel-2

Table 1. List of applicable documents

1.5.2. Reference documents

The following table lists the documents which contain information in support of this document.

Id.	Ref.	Description
RD1	https://doi.org/10.5281/zenodo.1209633	MAJA Algorithm Theoretical Basis Document.

RD2	Providing advice on Snow and Ice monitoring pre-implementation phase; Deliverable 4; Copernicus snow and ice Monitoring pre-implementation phase	EEA/ IDM/R0/18/006
RD3	Bippus G., Nagler T., Ripper E., Fernandes R., Metsämäki S. (2015): Methods and Protocols for Intercomparing and Validating Snow Extent and Snow Water Equivalent products – FINAL. Deliverable D7-SE.	SnowPex Deliverable 07
RD4	Cohen J. (1960): A coefficient of agreement for nominal scales. https://doi.org/10.1177/001316446002000104	

Table 2. List of reference documents

1.5.3. Auxiliary material

The following table lists the auxiliary data and the respectives url used in support of this document.

Id	Description and access
AUX1	Finnish Environmental Institute in situ data: https://www.syke.fi/fi-FI/Avoin_tieto/Ymparistotietojarjestelmat (last access: 05/05/2020)
AUX2	Hungarian Hydrological Data Centre in situ data: http://www.hydroinfo.hu/Html/archivum/archiv_jggraf.html (last access: 05/05/2020)
AUX3	EU-Hydro Upgrade User Guide v3.0, https://land.copernicus.eu/user-corner/technical-library/eu-hydro_v1-0_user_guide_d02.pdf (last access: 07/02/2020)

Table 3. List of auxiliary material

1.6. Terminology

The following table details abbreviations and acronyms that can be found in this document and the products files.

Abbreviations/acronyms	Description
AD	Applicable document
ATCOR	Atmospheric correction
AUX	Auxiliary material
CLMS	Copernicus Land Monitoring Service
HR-S&I	Pan-European High-Resolution Snow & Ice Monitoring of the CLMS
ICE	ICE processor (ASTRI POLSKA ice detection processor, containing the RLIE and ARLIE algorithms)
MACCS	Multisensor atmospheric correction and cloud-screening

MAJA	MACCS-ATCOR Joint Algorithm
NRT	Near real time
RD	Reference document
RLIE	River and Lake Ice Extent
S1	Sentinel-1
S1A	Sentinel-1A
S1B	Sentinel-1B
S2	Sentinel-2
S2A	Sentinel-2A
S2B	Sentinel-2B
VHR	Very High Resolution

Table 4. List of abbreviations and acronyms

2. Data and method

This section presents the quality assessment processes used for the RLIE S1 and RLIE S2 products. The overall summary of the validation of RLIE S1 and RLIE S2 products is presented in Figure 3.

As a reminder, RLIE products characterise the surface water areas defined by the EU-Hydro database [AUX3]. Both remote-sensing and in situ data collected on these areas are used for the assessment. For remote sensing information, both Sentinel-2 Level-1C (L1C) and MAJA Level-2A (L2A) products are studied. For more details on the MAJA software which performs atmospheric correction and cloud and cloud shadow detection, users should consult [AD3].

Readers should note that ice classification from S1 radar observations (RLIE S1 product) is only assessed on rivers. Classification over lakes is not yet mature enough (see the limitations described in [AD4], section 2.1.2.8.), and users are invited to use RLIE with caution on areas of standing water (refer to QCFLAGS layer).

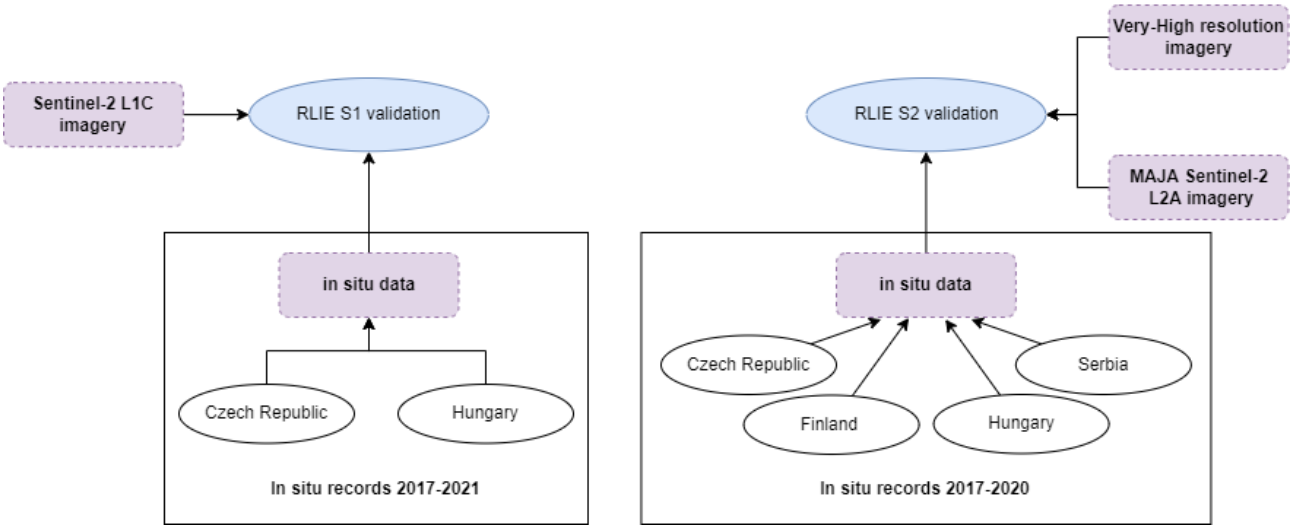


Figure 3. Overall summary of the HR-S&I ice products validation

2.1. Evaluation with in situ data

2.1.1. Data

In situ measurements were used for validation of the quality of ice products based on S1 and S2 data.

Requests for in situ data information were submitted to the meteorological and hydrometeorological institutions within EEA38+UK. Part of them do not conduct in situ measurements e.g. Denmark, Ireland, United Kingdom. Additionally, not all the institutions have replied to the request. Consequently, the countries where in situ measurements are collected were identified.

However, not all currently available data can be used for reliable validation of the RLIE product. In situ observations should include the geospatial localisation of the station, the observation date and the observed quantity (ice coverage) [RD2]. For some data, the coordinates of the station are not specified, which precludes any reliable geolocation of points of interest. Thus, much of the collected data are not applicable for the ice products quality assessment.

For the validation activities of RLIE S2 product, the in situ data from the following countries were used:

- Czech Republic
- Finland
- Hungary
- Serbia

For the evaluation of the RLIE S1 products, the in situ data from the following countries were used:

- Czech Republic
- Finland

The comparison of in situ measurements with RLIE S1 products was based on the in situ data records from the period of 2017-2021, and 2017-2020 for RLIE S2 product.

The in situ measurements are taken from water bodies, however their spatial representation is often pointed on land as they are made from stations located near dams. Therefore, all in situ data are manually checked to see if they are related to EU-Hydro polygon features.

Once the reliable stations for the evaluation (stations for which in situ data are complete and usable) are identified, the S1 and S2 acquisitions they correspond to are selected. In situ observations may not match S1 and S2 product dates. Indeed, for observations which are not conducted on a daily basis (e.g. 3 times a month) RLIE S2 data with 1-day time lag are considered. For RLIE S1 evaluation, only data acquired on the same date as in situ measurements are used. As the frequency of measurements widely varies between European countries and because there is no rule to collect in situ data on a daily basis, the set of the matching observations is considerably limited, especially in the case of RLIE S2 products where there are often clouds which exclude the data from the validation. In the case of the RLIE S1 products, only in situ measurements referring to rivers are taken into account for quality assessment. The performance of the ICE algorithm for RLIE S1 production is not assessed on lakes.

If a station is located on RLIE S1 or RLIE S2 overlapping tiles, no merging policy is applied. Tiles for this station are both taken into account and the in situ observation is processed for each tile.

For more information about the selected and available datasets and their characteristics, users should refer to Annex A.

2.1.2. Methodology

The whole methodology for processing in situ data is presented in Figure 4.

After the selection of in situ data, the information for each station is transformed into binary form (ice presence/no ice presence). When ice thickness information is given, measurements are converted to ice presence and absence using a threshold of 0 m. If the type of the ice cover is specified, observations corresponding to moving ice or any other ice types which are not usable for the evaluation are discarded (see Annex A for more details).

In situ data are used for quality assessment with the assumption that a single point observation is representative of a larger area corresponding to the pixel size of the RLIE

products [RD3]. This assumption is justified by the fact that in situ observations refer to the wide area of the inland water, although they are mainly conducted from land.

Consequently, during the in situ data processing, the classification from the RLIE referring to each in situ observation is assessed according to the following steps:

- an area with a radius of 100 m is defined around each point,
- only pixels located within both this area and the EU-HYDRO extent are considered,
- the point is assigned the RLIE class of the majority of the pixels.

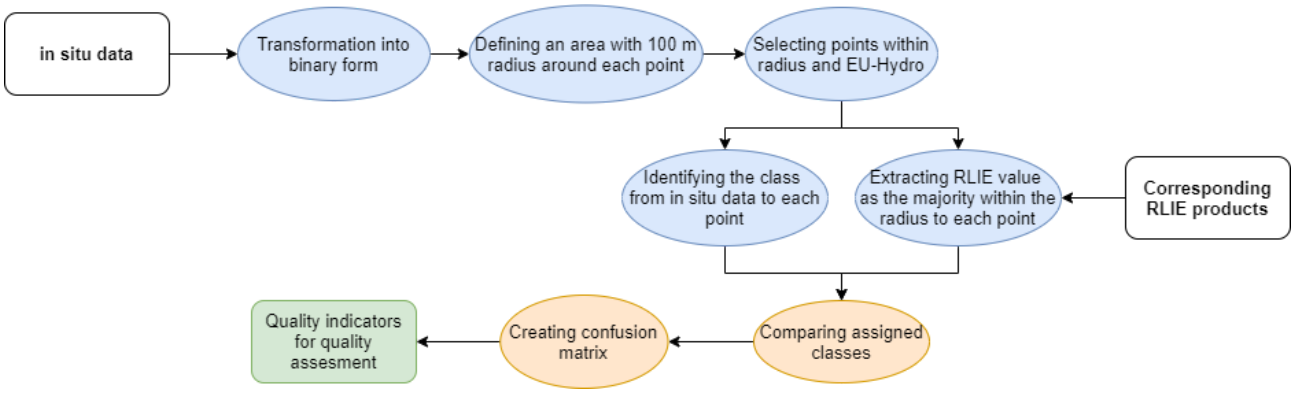


Figure 4. Workflow diagram for the quality assessment against in situ data

At each selected point, two class values are thus available: one resulting from the in situ measurement, the other from the RLIE S1 or RLIE S2 product. While all in-situ stations collect data regardless of weather conditions, RLIE S2 does not provide information for clouded parts of the images. Points for which the RLIE S2-derived class are 'clouds' or 'no data' are excluded from the validation process. RLIE S1 products based on S1 observations provide reliable information in clouded areas, and thus the number of observations to be compared is significantly greater than for RLIE S2 products.

Comparison results are divided into two groups:

- consistent results,
- erroneous results.

The first group (consistent results) refers to the observations for which ground-truth information (from in situ data) and RLIE class information are identical.

The group, defined as erroneous results, refers to observations for which RLIE-derived class is incorrectly assigned compared to in situ data.

For RLIE S1 and in situ data comparison, there are identical classes derived from both kinds of observations - in situ data and RLIE S1 product. In the case of RLIE S2 observations, there are three classes derived from RLIE S2: 'open water', 'snow-covered or snow-free ice', and 'other features', while only two identified classes from in situ data are 'open water' and 'snow-covered or snow-free ice'. Cases of RLIE S2 defined as 'other features' are also put in the group of 'erroneous results'.

Ultimately, a confusion matrix is created based on the consistent and erroneous results for RLIE S1 and RLIE S2 products separately.

2.2. Evaluation with remote-sensing data

2.2.1. RLIE S1 evaluation

2.2.1.1. Data

Sentinel-2 L1C images

The quality assessment of the RLIE S1 product is performed based on the comparison of the manual classification results of the corresponding Sentinel-2 L1C imagery.

S2 L1C products are used for the evaluation purposes of the RLIE S1 products. S2 satellites have a Multi-Spectral Instrument (MSI) with 13 spectral bands that vary in spatial resolution: 4 spectral bands at 10 m resolution (B2 - Blue, B3 - Green, B4 - Red, B8 - NIR), 6 spectral bands at 20 m resolution (B5, B6, B7, B8A, B11, B12) and 3 spectral bands at 60 m resolution (B1, B9, B10). For the purpose of the manual classification of ice cover on S2 L1C imagery, the bands at 10 m spectral resolution are used. Consequently, the validation of the RLIE S1 product (delivered at 20 m resolution) is based on the comparison with satellite data with a higher spatial resolution (namely 10 m).

For the purpose of those validation activities, S2 data from different time periods were chosen. Data with the ice cover occurrence were taken into account. In order to ensure a comparison of the same ice phenomena within a particular water body between S1 and S2 images, only satellite images acquired on the same day as RLIE S1 data were chosen for the validation to ensure the reliability of such comparisons. However, users should note that the time of S1 and S2 data acquisition varies, and consider that the ice phenomenon can change within this time lag, especially because of the flowing or moving ice occuring on rivers.

The cloud coverage in S2 images was checked visually and from the metadata of the archive images. Only cloud-free or nearly cloud-free images within the extent of inland waters were kept.

Due to frequent cloud coverage in winter, the set of S2 data meeting the validation requirements is restricted. Ultimately, a dataset of S2 images provided below from different EEA38+UK sites was chosen for the quality assessment of the RLIE S1 product. Their location is presented in Figure 5 and detailed in Annex B (see Table 23). Given the spatial and temporal distribution of the selected images, the S2 images dataset is considered representative.

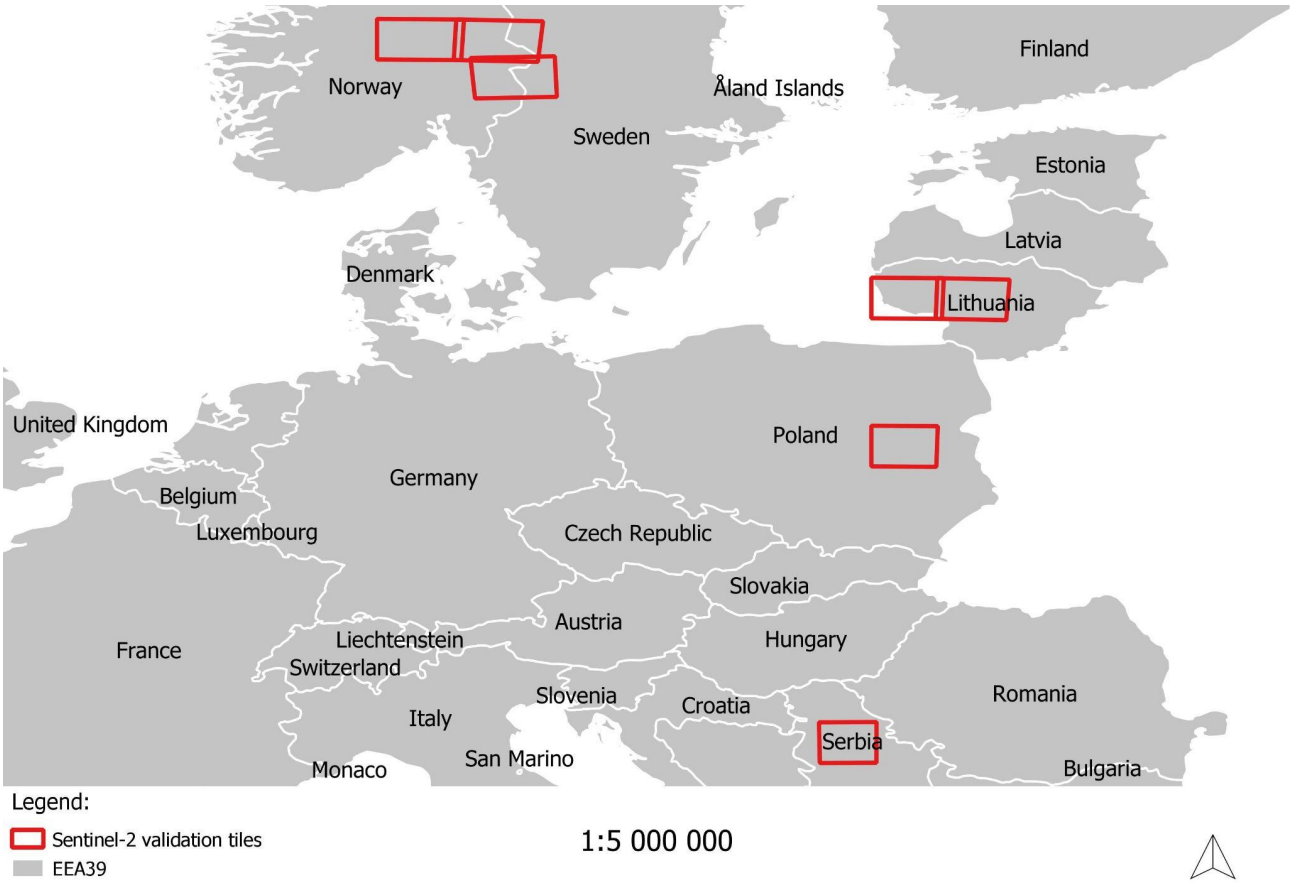


Figure 5. Localisation of the Sentinel-2 tiles used for the evaluation of the ice detection accuracy in RLIE S1 product.

2.2.1.2. Methodology

Sentinel-2 L1C imagery

S2 L1C images are acquired as orthorectified products, and the pre-processing of the images is not required. For the scope of the validation, S2 L1C RGB compositions are generated and used for manual classification.

First, a regular point grid is generated within the EU-Hydro polygons. The spacing of points is no less than 20 m. The value is based on RLIE S1 spatial resolution and ensures that generated points refer to different RLIE S1 pixels. These points are recognised as accuracy assessment points.

Secondly, for the accuracy assessment points, the river/lake coverage is determined based on the visual classification of S2 data. The points were assigned to one of the following RLIE S1 coverage classes:

- open water
- snow-covered or snow-free ice

The identification of the river/lake coverage is based on the experience and knowledge of ice experts. Clouded parts of S2 images are visually detected and the associated accuracy assessment points are excluded from the evaluation.

Thirdly, the coverage class from the RLIE S1 product is assigned to the same set of accuracy assessment points. Points for which the RLIE S1 class is defined as 'no data' are excluded from the comparison. In the case of overlapping RLIE S1 tiles, the class from all tiles are taken into account and no merging policy is applied. As a result, points from the overlapping areas are duplicated and assigned to all available RLIE S1 classes.

Generating a regular grid and assigning the S2 class to each grid point by manual interpretation allows intercomparison without resampling the S2 images to a coarser grid (matching the spatial resolution of RLIE S1 products). Moreover, the key advantage of this approach comes from the fact that manual interpretation of the river/lake coverage in S2 images at 10 m resolution is more reliable (lower error) than any automated classification algorithm.

2.2.2. RLIE S2 evaluation

2.2.2.1. Data

VHR images

The quality assessment of the RLIE S2 product is performed based on VHR optical remote-sensing images from SPOT 6/7 and Pléiades 1A/1B.

SPOT-6 and SPOT-7 images have 1.5 m panchromatic resolution and 6 m multispectral (blue, green, red, near-infrared) resolution. Pléiades 1A/1B images are available at finer 0.5 m panchromatic resolution and 2 m multispectral (blue, green, red, near-infrared) resolution. Sensors acquire images with a large radiometric depth coded in 16 bits, which is a key advantage for ice/snow detection. Data are retrieved as orthorectified products from Airbus Defence and Space.

SPOT 6/7 and Pléiades 1A/1B images were analysed in the EU-Hydro extent [AUX3] over the following periods:

- January 2017 - April 2017,
- November 2017 - April 2018,
- November 2018 - April 2019,
- November 2019 - January 2020.

In order to ensure a comparison of the same ice phenomena within a particular water body between S2 and VHR images, only satellite images with immobile ice cover were considered, and the selected images were checked to not include constantly moving floe. The selection was made visually, as the floe can be easily detected by an experienced image interpreter. Moreover, in order to provide the reliability of such comparisons, a threshold of one-day time lag is applied when selecting S2 and VHR images.

The cloud coverage in VHR images was checked visually and from the metadata of the archive images. Only cloud-free or nearly cloud-free images within the extent of inland waters were kept.

Due to frequent cloud coverage in winter, only a few pairs of S2 and VHR images meeting the validation requirements were identified. Ultimately, a dataset composed of six images from different EEA38+UK sites was chosen for the quality assessment of the RLIE S2 product. Their location is presented in Figure 6 and detailed in Annex B (see Table 21).

The VHR images dataset is checked to be representative of different ice types, as follows:

- snow-covered ice,
- snow-free ice,

- particularly thin ice,
- broken ice,
- partially ice cover.

Those various ice forms can be visually differentiated, as shown by the examples in Annex C.

Given the spatial and temporal distribution of the selected images and the various ice types provided, the VHR images dataset is considered representative.

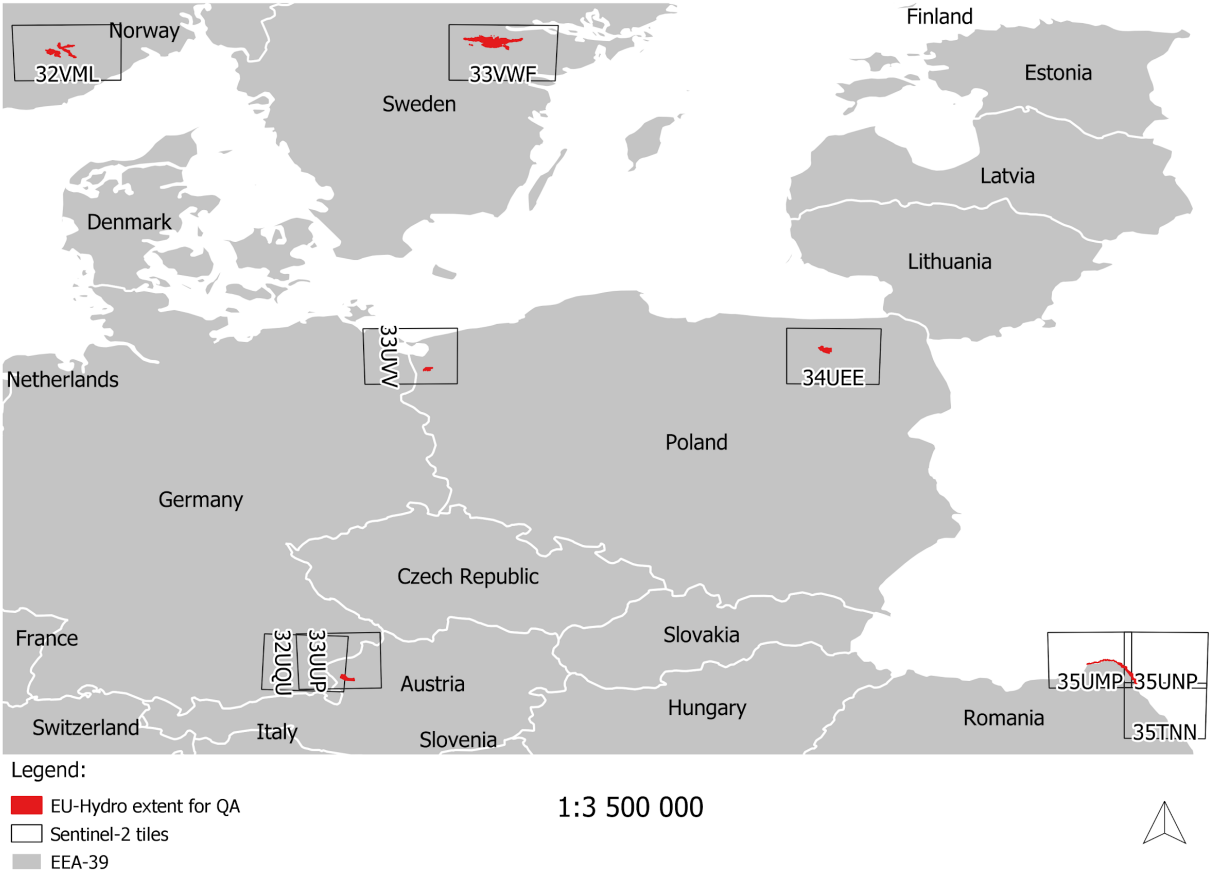


Figure 6. Localisation of the EU-Hydro features used for the evaluation of the ice detection accuracy and corresponding Sentinel-2 tiles.

MAJA Sentinel-2 L2A images

An additional validation against MAJA S2 L2A images was performed. The L2A products were selected in order to particularly assess the RLIE S2 products on water bodies affected by turbidity and water bloom or covered by thin/dark ice, as the ICE algorithm presents misclassification issues in these areas (see the limitations section in [AD3]).

Finally, 9 MAJA L2A images were chosen at different periods of the year in various locations detailed in Annex B (see Table 22) and validation is performed on the RGB compositions at 10 m resolution.

2.2.2.2. Methodology

VHR images

VHR images are acquired as orthorectified products and the pre-processing of the images is not required.

First, a regular 20 m grid is generated within the EU-Hydro polygons. The spacing of 20 m is based on S2 resolution and ensures that generated points refer to an RLIE S2 pixel. These points are recognised as accuracy assessment points.

Secondly, polygons referring to one of the following RLIE coverage classes are manually drawn on the VHR images:

- open water
- snow-covered or snow-free ice
- other features

The identification of these polygons is based on the experience and knowledge of a lake and river ice expert. Clouded parts of VHR images are visually detected and the associated accuracy assessment points are excluded from the evaluation.

Thirdly, the coverage class from the RLIE S2 product is assigned to the same set of accuracy assessment points. Points for which the RLIE S2 class is defined as 'clouds' and 'no data' are excluded from the comparison. In the case of overlapping RLIE S2 tiles, the class from all tiles are taken into account and no merging policy is applied. As a result, points from the overlapping areas are duplicated and assigned to all available RLIE S2 classes.

Generating a regular grid and assigning the VHR class to each grid point by manual interpretation allows intercomparison without resampling the VHR images to a coarser grid. Moreover, the key advantage of this approach comes from the fact that manual interpretation of the river/lake coverage in VHR images at 0.5 and 1.5 m resolution is more reliable (lower error) than any automated classification algorithm.

MAJA Sentinel-2 L2A images

For the quality assessment against MAJA S2 L2A images, a methodology similar to that used for the VHR images is applied. Only two steps differ:

- clouded pixels are excluded from the validation using the cloud mask provided in the MAJA L2A product, instead of visual identification of the clouds for VHR images
- polygons and accuracy assessment points are divided into two groups: (1) points and polygons referring to lakes (2) points and polygons referring to rivers

Two confusion matrices are computed from the accuracy assessment points to evaluate the ICE algorithm performance on (1) lakes and on (2) rivers.

2.3. Evaluation metrics

The confusion matrix summarises the results from the validation of the ICE algorithm against reference data (in situ data, VHR images, MAJA S2 L2A images, S2 L1C imagery).

Once the thematic classes from the reference data and RLIE products are determined for each accuracy assessment point, intercomparison results are gathered in a confusion matrix (Figure 7). Eventually, confusion matrices are generated for the quality assessment against in

situ data, VHR images, S2 L2A images (with a distinction of the results between rivers and lakes) and S1 L1C products.

The main diagonal (green) of the matrix contains the cases where the ICE algorithm classification and reference dataset agree, whereas the off-diagonal elements (red) contain the disagreement cases.

RLIE / VHR	open water	snow-covered or snow-free ice	other features
open water	n_{11}	n_{12}	n_{13}
snow-covered or snow-free ice	n_{21}	n_{22}	n_{23}
other features	n_{31}	n_{32}	n_{33}

Figure 7. Confusion matrix definition (applicable for RLIE S2 assessment only as RLIE S1 product does not include the "other features" class)

The following meaningful quality indicators are derived from each confusion matrix:

- overall accuracy
- commission error
- omission error
- precision
- recall
- F-measure
- kappa coefficient [RD4]

The calculation of these quality indicators is detailed in Annex D.

3. Results

This section gathers the evaluation results of the RLIE S1 and RLIE S2 products based on in situ data and optical remote-sensing images.

For details concerning the quality assessment methodology, users should refer to section 2.

3.1. Evaluation with in situ data

3.1.1. RLIE S1

RLIE S1 products having corresponding in situ measurements recognised as reliable are used for validation purposes. Classifications given by in situ measurements (from the Czech Republic, and Hungary) and matching RLIE S1 products are tabulated in one confusion matrix (see Table 5). The derived quality indicators based on the confusion matrix are presented in Table 6.

On the whole, 18 101 observations are used for the validation. Classification of the rivers and lakes coverage (open water or snow-free/snow-covered ice) from in situ measurements and the RLIE S1 product is consistent for 13 882 points. The overall accuracy of the ICE algorithm for RLIE S1 production derived from the evaluation against in situ data is therefore 77%. Considering the evaluation metrics (see Table 6) the omission and commission error values are high for the class 'snow-covered or snow-free ice' and only 25% of ice cover is correctly detected. Based on the recall and precision values, only 24.9% of the ice cover is discriminated, and only 38.6% of the detected ice is actually ice cover.

The identified causes of ICE algorithm misclassification of the ice cover comparing to in situ measurements are related to:

1) ICE algorithm classification limitations (for more detail please refer to the limitations section in [AD4]) - All of the ICE algorithm limitations significantly decrease its overall accuracy. Thus, before using the RLIE S1 products, users are asked to get familiar with those limitations and often consider the RLIE S1 products with caution. We also invite users to use RLIE S1 products together with its QC layer - which can be helpful to assess the areas where the classification results are of low reliability (for more information about the QC layer please refer to [AD2]).
2) in situ observations:
    a) uncertainty of the ice observations - ice observations are conducted by ice observers and the reliability of this data is not provided by the relevant institutions
    b) geolocation of the points - geographical location of the observation points is approximate. To mitigate this lack of precision, the RLIE S1 classifications in the radius of 100 m around the point are taken into account (see section 2.1.2). However, as some in situ points are located outside the EU-Hydro extent, the compared data can refer to river banks and thus it can decrease the overall accuracy of the ICE algorithm.
    c) Location of in situ stations near dams and other artificial objects - as presented on Figure 9, Figure 10, and Figure 11 ice measurements are very often conducted near dams and other artificial objects (bridges etc.). The ICE

algorithm can not distinguish artificial objects as a result of the SAR sensors limitations. In most of the situations dams and other artificial objects are classified as 'snow-covered or snow-free ice', it therefore can significantly decrease the algorithm accuracy.

d) in situ measurements located on rivers/river parts which are narrower than 150 m - in situ points are also often located on narrow river parts (see Figure 8). Taking into account ICE algorithm limitations [AD4] the classification results on those objects can be inaccurate.

e) ice type is not defined - for Czech Republic the ice cover value is only provided without the discrimination of the ice type. Taking that into account, ice values can concern border ice, flowing ice etc. This fact can also influence the disagreement between RLIE S1 and in situ cover.

in situ measurements	RLIE S1	
	open water	snow-covered or snow-free ice
Open water	12966 (72%)	1454 (8%)
Snow-covered or snow-free ice	2765 (15%)	916 (5%)

Table 5. Confusion matrix between RLIE S1 product and in situ measurements

Quality indicator	RLIE S1 class	
	open water	snow-covered or snow-free ice
Overall accuracy [%]	**76,7**	
Kappa coefficient [%]	**17,1**	
Omission error [%]	10,1	75,1
Commision error [%]	17,6	61,4
Recall [%]	89,9	24,9
Precision [%]	82,4	38,6
F1 score [%]	86,0	30,3

Table 6. Results of the validation of RLIE S1 products with in situ measurements

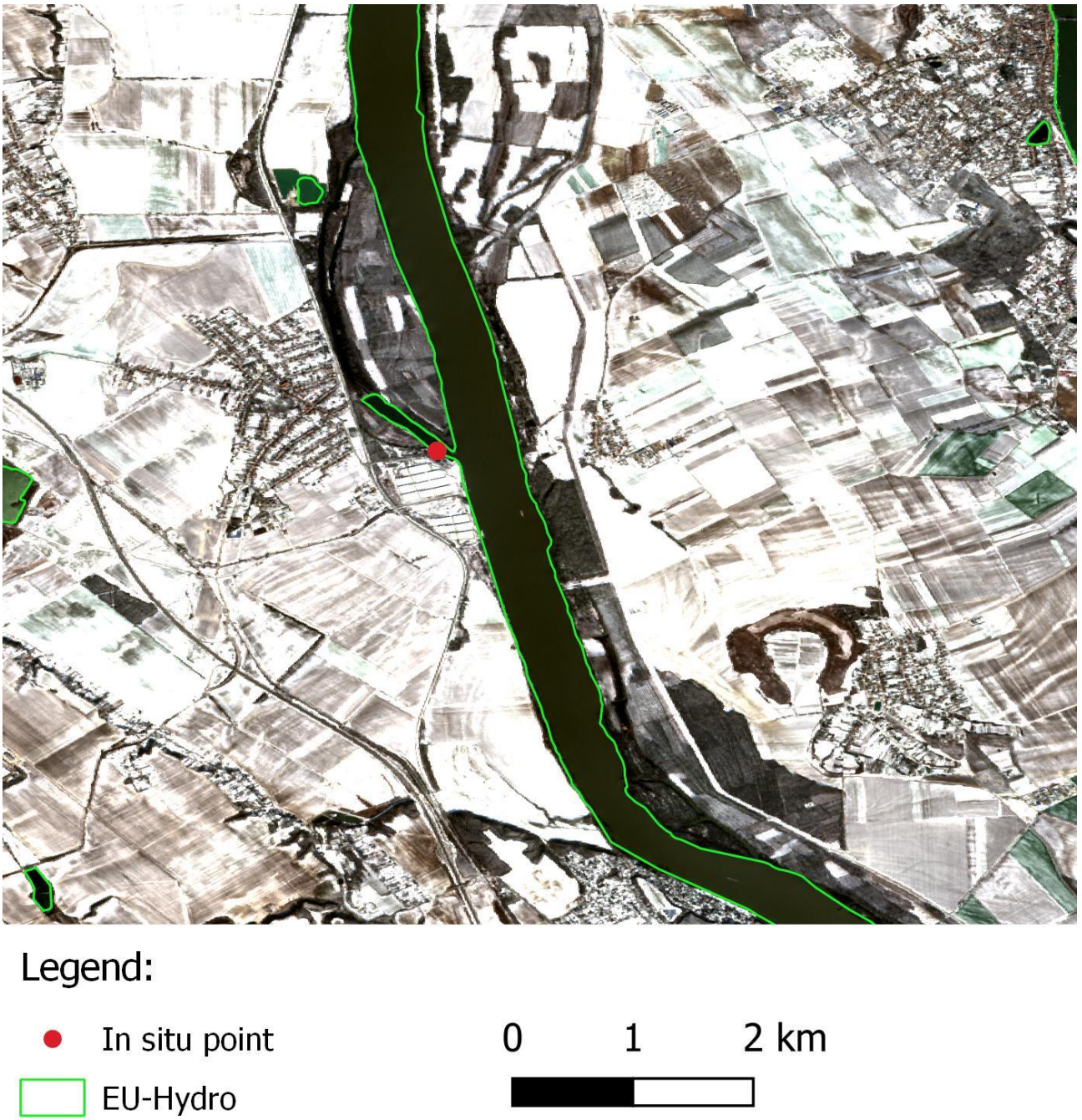


Figure 8. In situ point located on the narrow part of the river. Background: Sentinel-2 RGB T33TYN 13.02.2021

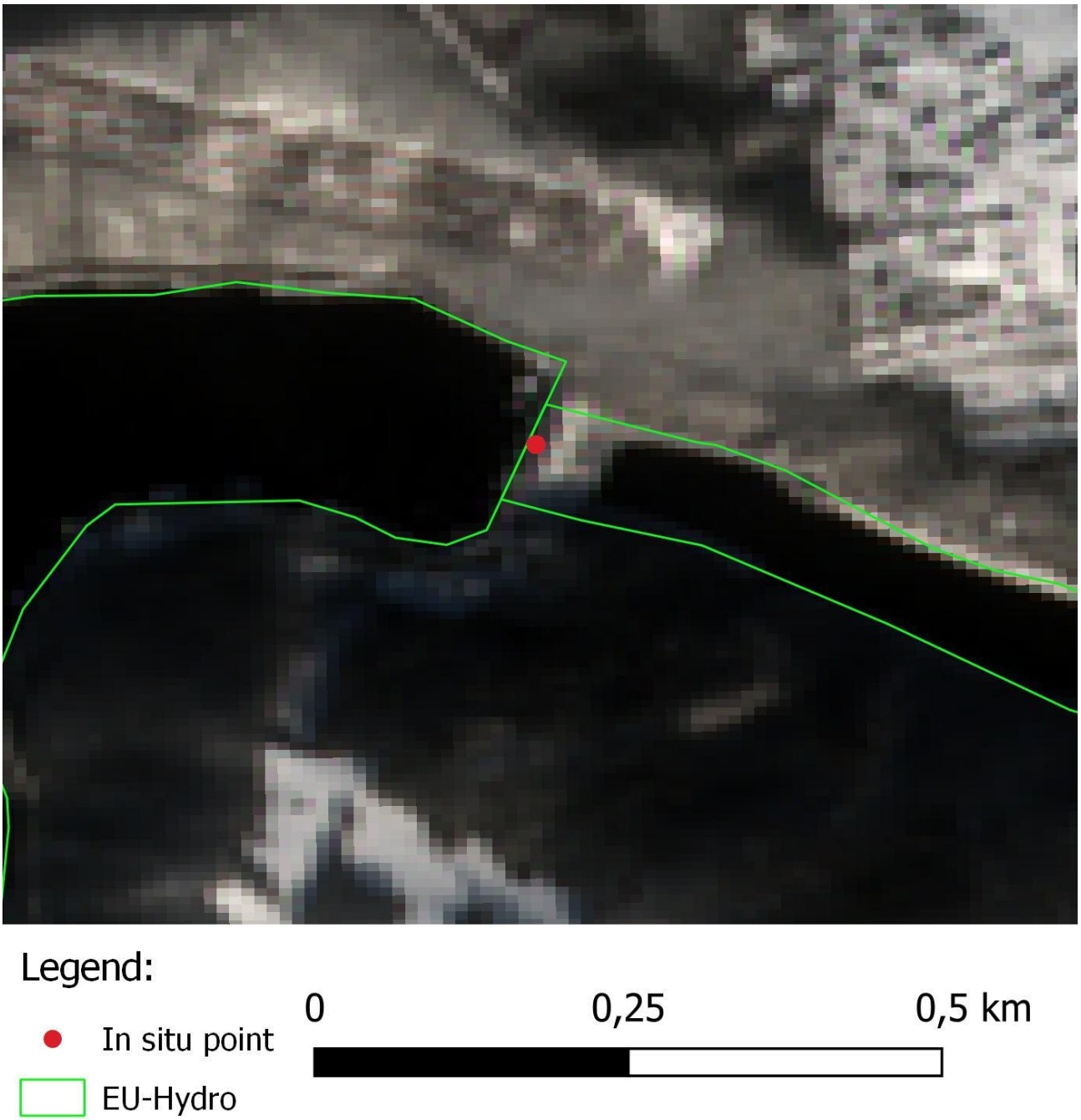


Figure 9. In situ point located near the dam. Background: Sentinel-2 RGB T33UVR, 12.02.2021

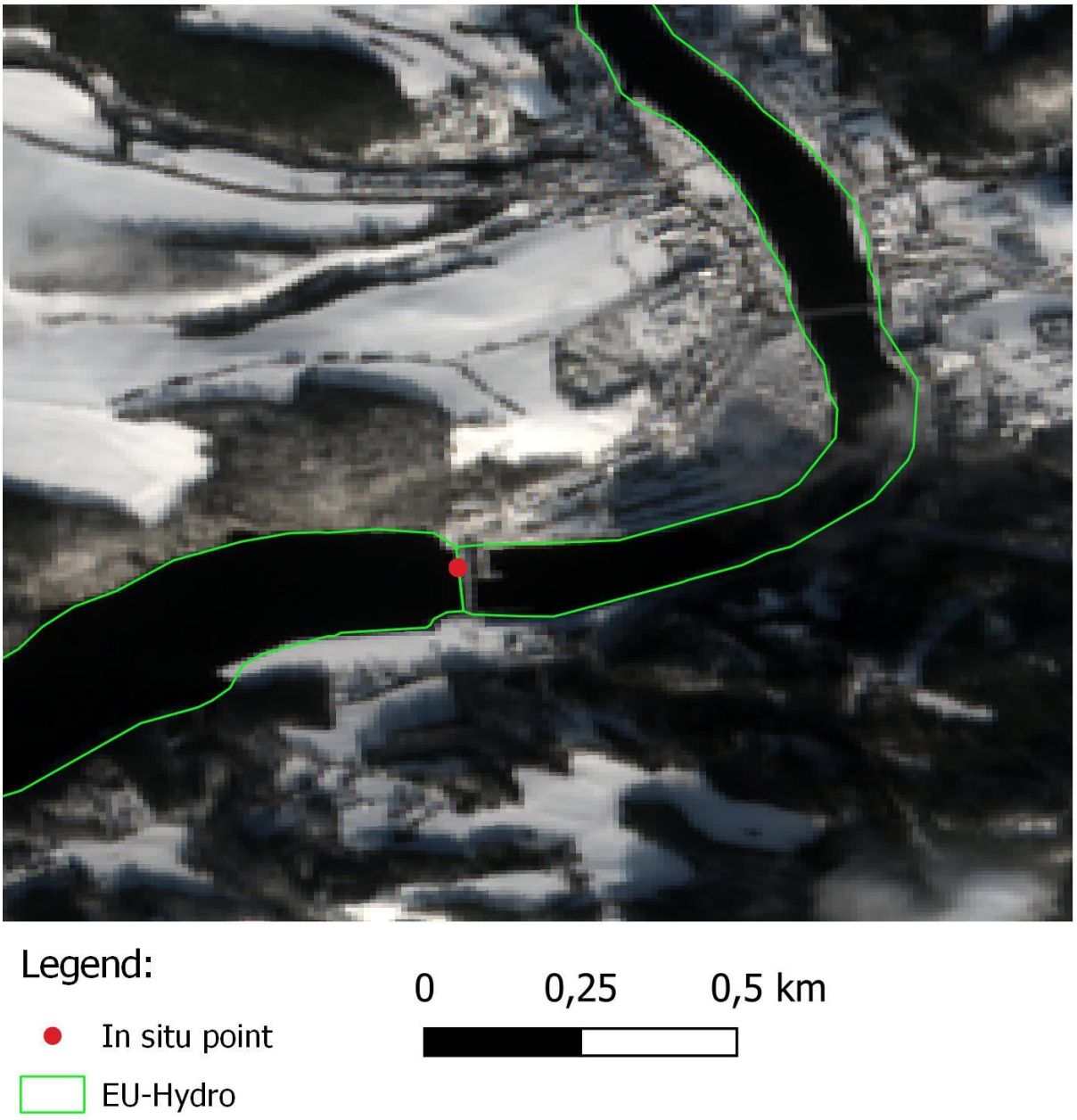


Figure 10. In situ point located near the bridge. Background: Sentinel-2 RGB T33UVR, 12.02.2021

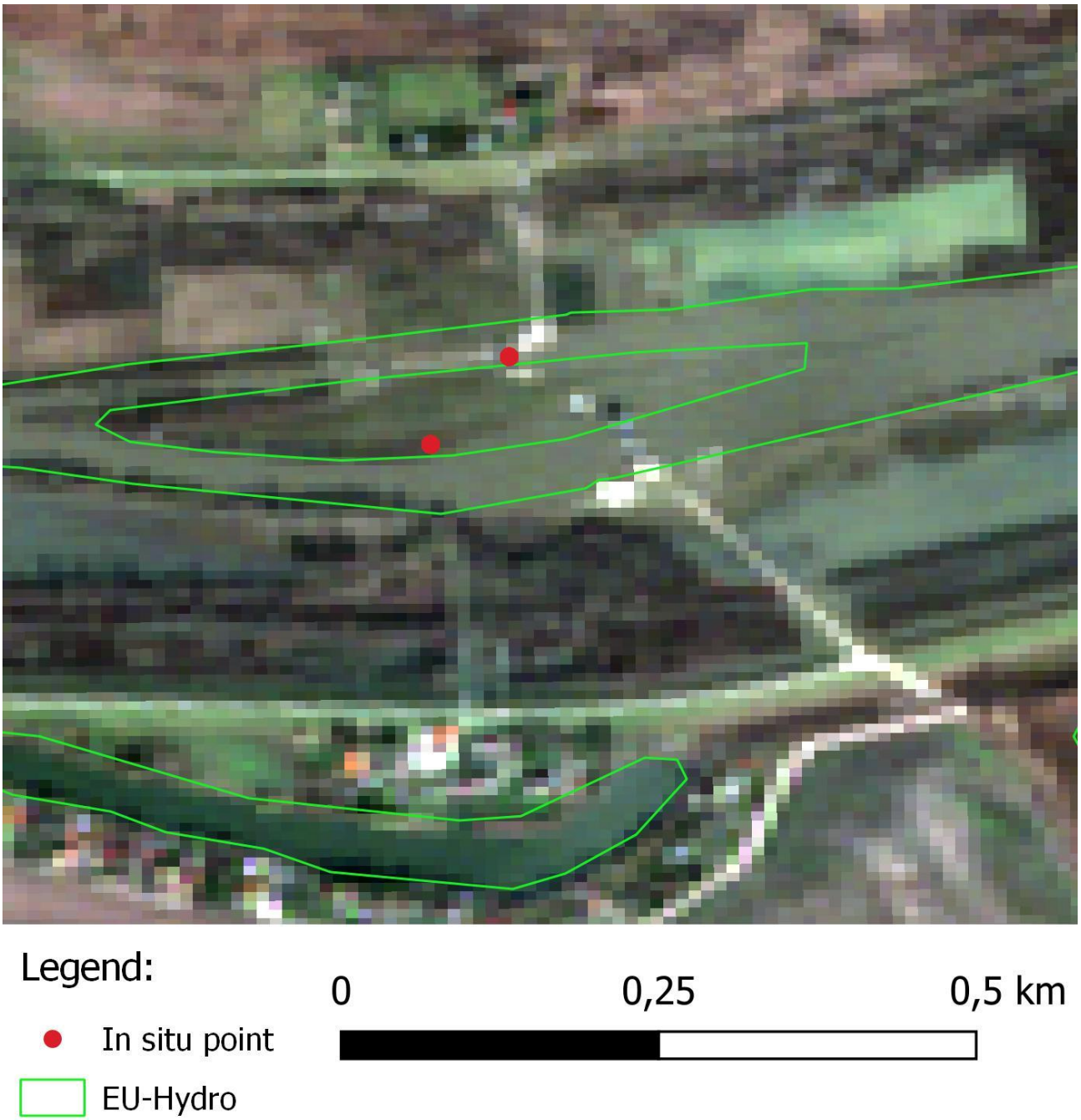


Figure 11. In situ points located near artificial objects. Background: Sentinel-2 RGB T34TDS, 28.02.2021, Koros river, Hungary.

3.1.2. RLIE S2

In situ measurements recognised as reliable and having corresponding cloudless RLIE S2 products are used for validation purposes. All in situ measurements (from the Czech Republic, Finland, Hungary, and Serbia) and corresponding RLIE S2 products' classification are tabulated

in one confusion matrix (see Table 7). The derived quality indicators based on the confusion matrix are presented in Table 8.

in situ measurements	RLIE S2		
	open water	snow-covered or snow-free ice	other features
Open water	483 (64%)	4 (0,5%)	13 (1,7%)
Snow-covered or snow-free ice	91 (12%)	162 (21%)	4 (0,5%)
Other features	0	0	0

Table 7. Confusion matrix between RLIE S2 product and in situ measurements

Quality indicator	RLIE S2 class		
	open water	snow-covered or snow-free ice	other features
Overall accuracy [%]	**85,2**		
Kappa coefficient [%]	**65,2**		
Omission error [%]	3,4	37,0	0,0
Commision error [%]	15,9	2,4	100,0
Recall [%]	96,6	63,0	0,0
Precision [%]	84,1	97,6	0,0
F1 score [%]	89,9	76,6	0,0

Table 8. Results of the validation of RLIE S2 products with in situ measurements

On the whole, 757 observations are used for validation purposes. The classification from in situ measurements and the RLIE S2 product is consistent for 645 points. The overall accuracy of the ICE algorithm derived from the evaluation against in situ data is therefore 85% and the kappa coefficient is 65%. Considering the evaluation metrics (see Table 8) the omission error value is high - 37% - for the class 'snow-covered or snow-free ice'. The cause of ICE algorithm misclassification of the ice cover comparing to in situ measurements can be the fact that the ice coverage collected by in situ measurements can refer to the thin ice which discrimination is a known ICE algorithm limitation (for more detail please refer to the limitations section in [AD3]). Taking into account the ice type is not defined for all used in situ datasets (e.g. Czech Republic), the ice coverage derived from in situ measurements can be also assigned with border ice or moving ice which is hardly detected by the ICE algorithm (for more detail please refer to the limitations section in [AD3]). As in situ data always refer to 'open water' or 'snow-covered or snow-free ice', the ICE algorithm accuracy is decreased by the cases where RLIE S2 derived class is 'other features' and resulting from that the commission error for 'other

features' is 100%. Nevertheless, in situ measurements are always conducted near dams and other artificial object [see Figure 9, Figure 10, and Figure 11] (which are to be classified as 'other features' by the ICE algorithm) and consequently the predominant class in the radius of 100 m derived from RLIE S2 products is probable to be 'other features' class (especially when the dam is located on a narrow river). Considering all the aforementioned misclassification issues, the ICE algorithm for RLIE S2 overall accuracy calculated based on the validation against in situ measurements is still on the required level of 85%.

3.2. Evaluation with remote-sensing data

3.2.1. RLIE S1 evaluation results

Sentinel-2 L1C images

The evaluation of the RLIE S1 ICE algorithm is done against the set of manually interpreted S2 L1C images. The confusion matrix between the evaluation data (S2 imagery) and corresponding RLIE S1 products is given in Table 9. The derived quality indicators based on the confusion matrix are presented in Table 10.

Sentinel-2 class	RLIE S1	
	open water	snow-covered or snow-free ice
Open water	23101 (19%)	29729 (24%)
Snow-covered or snow-free ice	1281 (1%)	69499 (56%)

Table 9. Confusion matrix between RLIE S1 product and Sentinel-2 images

Quality indicator	RLIE S1 class	
	open water	snow-covered or snow-free ice
Overall accuracy [%]	**74,9**	
Kappa coefficient [%]	**45,0**	
Omission error [%]	5,3	30,0
Commision error [%]	56,3	1,8
Recall [%]	94,7	70,0
Precision [%]	43,7	98,2

F1 score [%]	59,8	81,8

Table 10. Results of the validation of RLIE S1 products with Sentinel-2 images

The confusion matrix comparing the RLIE S1 products and the S2 images shows that the ICE algorithm's overall accuracy is above the level of 74,9% and the kappa coefficient is at 55,0%. According to the evaluation metrics, 98.2% of the classified ice pixels is actually ice, however only 43,7 of the classified 'open water pixels is actually water cover. The 'snow-covered or snow-free ice' class has low commission error and average omission error. The commission error for the class 'open water' is 56,3%. Analysis of these errors led to the conclusion that the ICE algorithm for RLIE S1 production is prone to underestimating ice coverage. It is caused by the ICE algorithm limitations resulting directly from the limitations of the SAR sensors like S1. All the limitations of the ICE algorithm are detailed in [AD4]. The main restriction of the S1 imagery significantly influencing on the validation results is the discrimination of perfectly smooth ice coverage, wet-snow covered ice surface and thin ice - these ice forms are prone to be misclassified as 'open water'. In Table 11 the example of the water class overestimation is presented. In this example from Glomma river in Norway, the ICE algorithm does not distinguish smooth ice cover correctly. Lots of the ice pixels are classified as 'open water'. Table 12 presents the example of Sava river in Serbia where the thin border ice cover can be seen on S2 RGB composition. Here, the ICE algorithm does not classify areas covered with thin ice as the results of its known limitations. The last example provided in Table 13 is from the Havel river in Germany. In this example the huge underestimation of ice cover can be seen. The areas covered with snow are not recognised by the ICE algorithm. Please note that in this example the characteristic corridor of flowing ice on the river can be seen and it is quite well discriminated by the ICE algorithm. It proves that ICE algorithm is prone to recognise well flowing ice which can form ice jams on rivers which can threaten people and local properties.

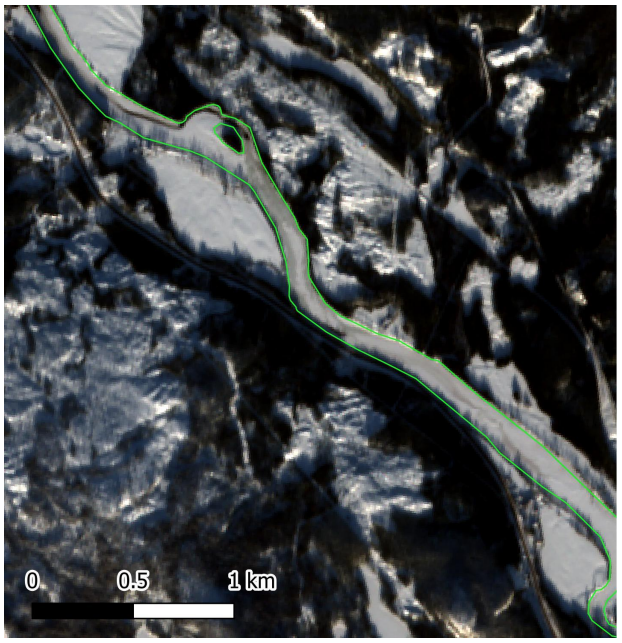
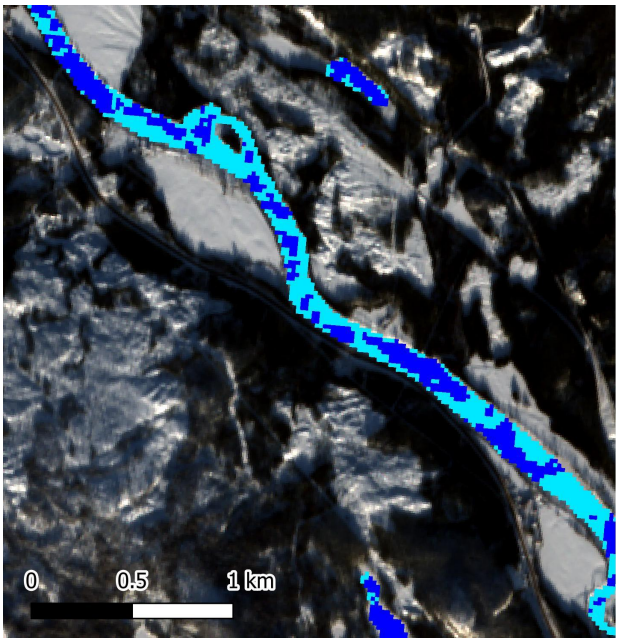
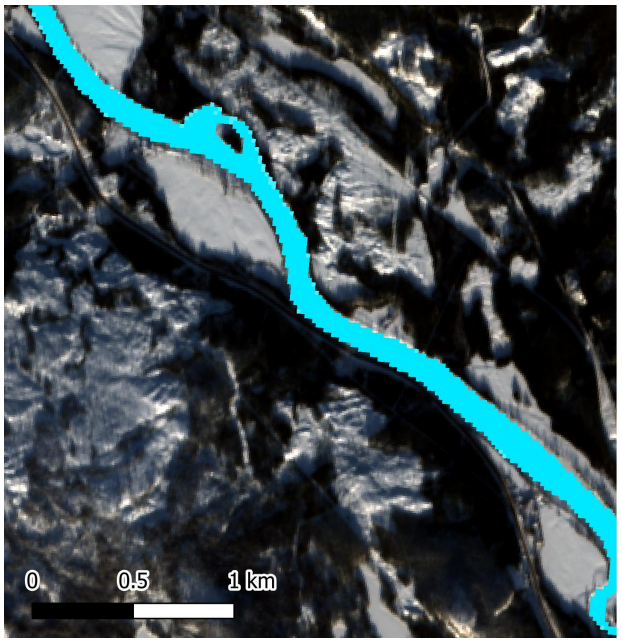
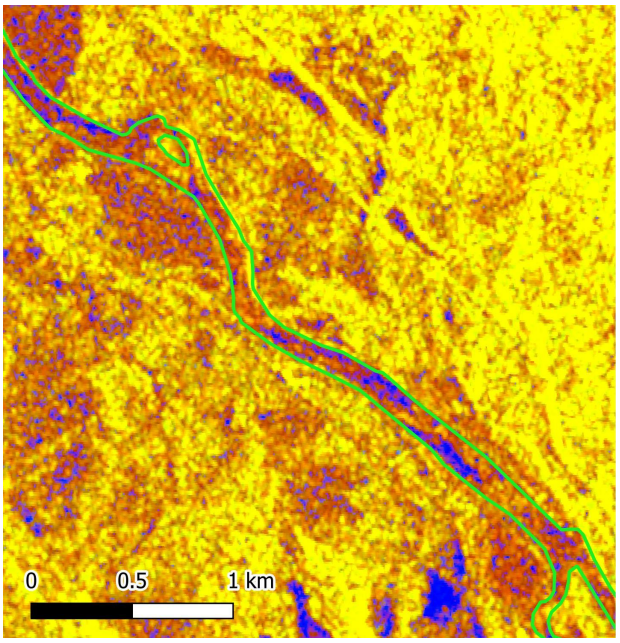
	
Sentinel-2 RGB composition 04.02.2021; Norway, Glomma river	RLIE S1 product 04.02.2021; Norway, Glomma river
	
Classified Sentinel-2 product 04.02.2021; Norway, Glomma river	False colour Sentinel-1 composition 04.02.2021; Norway, Glomma river

Table 11. Example of the ICE algorithm misclassification of smooth ice cover on rivers. Snow-covered or snow-free ice (cyan), open water (blue). Green lines delimit EU-Hydro extent.

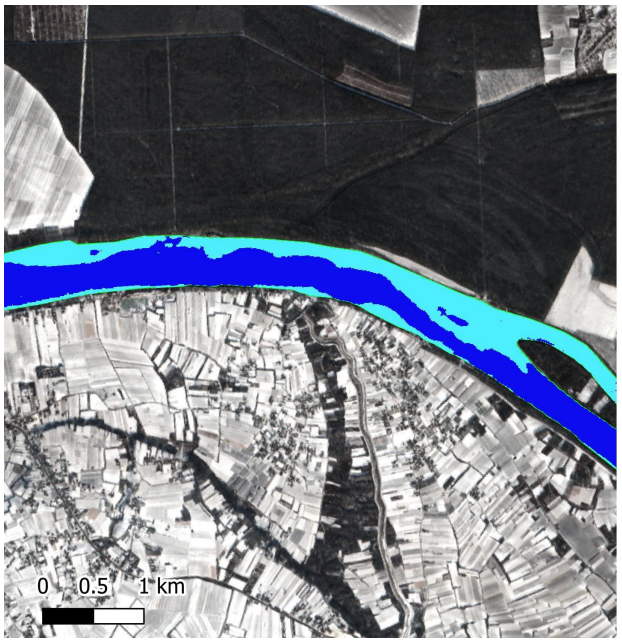
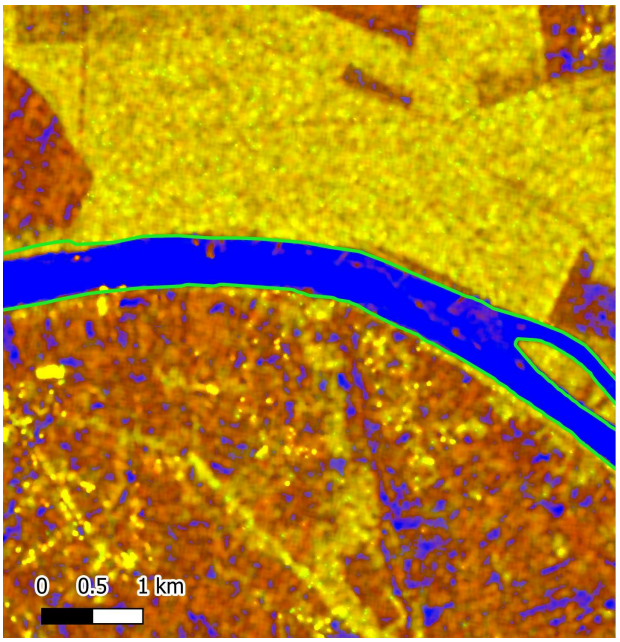
Sentinel-2 RGB composition 15.01.2017; Serbia, Sava river	RLIE S1 product 15.01.2017; Serbia, Sava river
	
Classified Sentinel-2 product 15.01.2017; Serbia, Sava river	False colour Sentinel-1 composition 15.01.2017; Serbia, Sava river

Table 12. Examples of the ICE algorithm misclassification of thin ice cover on rivers. Snow-covered or snow-free ice (cyan), open water (blue). Green lines delimit EU-Hydro extent.

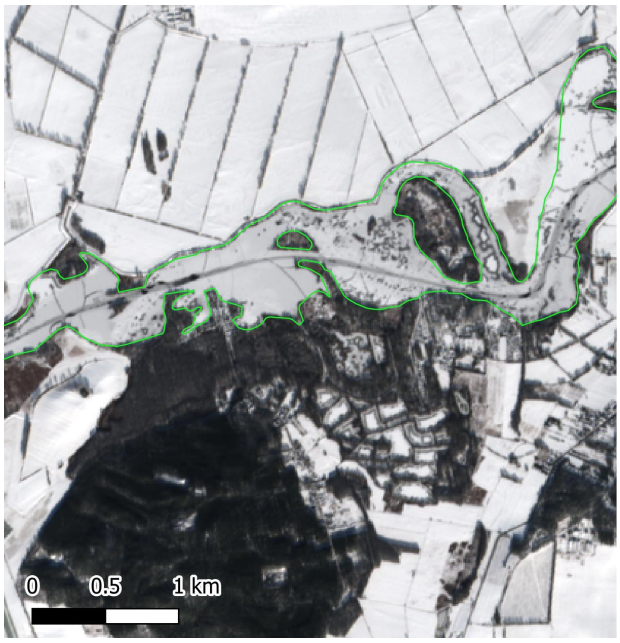
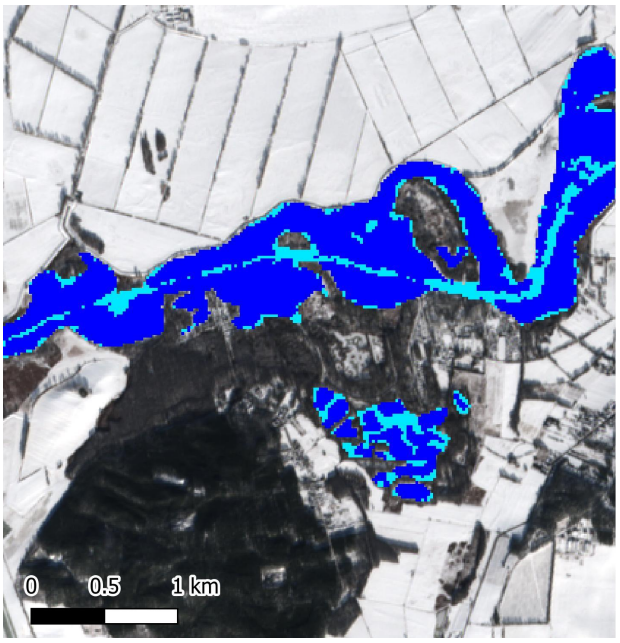
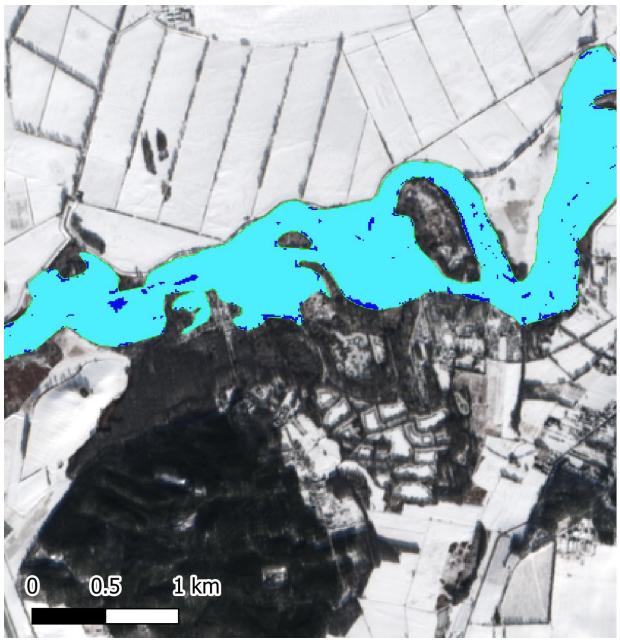
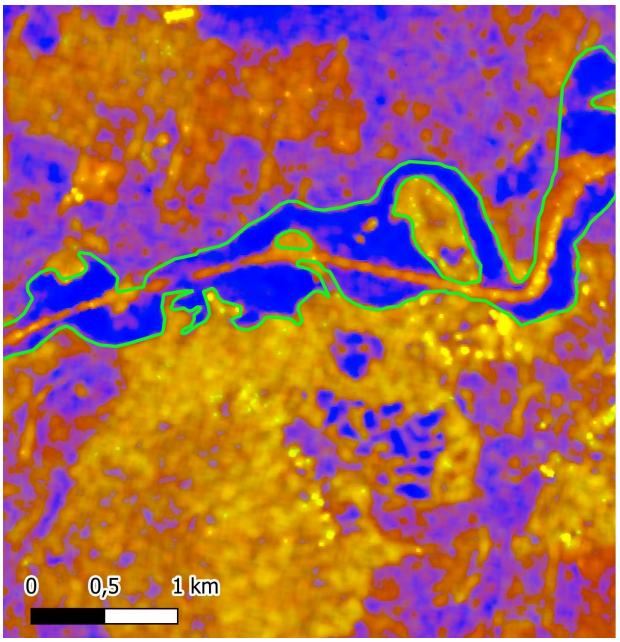
	
Sentinel-2 RGB composition 12.02.2021; Germany, Havel river	RLIE S1 product 12.02.2021; Germany, Havel river
	
Classified Sentinel-2 product 12.02.2021; Germany, Havel river	False colour Sentinel-1 composition 12.02.2021; Germany, Havel river

Table 13. Examples of the ICE algorithm misclassification of snow-covered ice on rivers. Snow-covered or snow-free ice (cyan), open water (blue). Green lines delimit EU-Hydro extent.

3.2.2. RLIE S2 evaluation results

VHR images

The evaluation of the ICE algorithm is done against six manually interpreted SPOT 6/7 and Pléiades 1A/1B images. The confusion matrix between these six VHR images and corresponding S2 images is given in Table 14. The derived quality indicators based on the confusion matrix are presented in Table 15.

VHR class	RLIE S2		
	open water	snow-covered or snow-free ice	other features
Open water	40898 (24%)	220 (<1%)	7 (<1%)
Snow-covered or snow-free ice	7105 (4%)	119757 (71%)	115 (<1%)
Other features	88 (<1%)	51 (<1%)	677 (<1%)

Table 14. Confusion matrix between RLIE S2 product and VHR images

Quality indicator	RLIE S2 class		
	open water	snow-covered or snow-free ice	other features
Overall accuracy [%]	**95.5**		
Kappa coefficient [%]	**88.7**		
Omission error [%]	0.6	5.7	17.0
Commision error [%]	15.0	0.2	15.3
Recall [%]	99.4	94.3	83.0
Precision [%]	85.0	99.8	84.7
F1 score [%]	91.7	97.0	83.8

Table 15. Results of the validation of RLIE S2 products with VHR image

The confusion matrix comparing the RLIE S2 products and the VHR images shows that the ICE algorithm's overall accuracy is above the required level of 85%. The overall accuracy (proportion of correct classifications) of the ICE results is about 96% and the kappa coefficient is at 89%. This proves that the ICE algorithm for RLIE S2 production provides good-quality results and shall be considered as a reliable ice classification tool. According to the evaluation

metrics, the 'snow-covered or snow-free ice' class has low omission and commission errors. The commission error for the class 'open water' is 15%. Analysis of these errors led to the conclusion that the ICE algorithm is prone to underestimating ice coverage when ice is extremely thin (see Table 16). This limitation is also described in [AD3]. It was observed that the issue often occurs with partially ice-covered water bodies. In these cases, misclassification is explained by the nearly same spectral reflectances of ice and water. The omission and commission errors for class 'other features' (vegetation and soil) are respectively 17% and 15.3%. However the distinction of other features (vegetation and soil) is challenging as features in this class can have considerably different reflectances, e.g. due to various vegetation conditions and different soil types. 'Other features' are thus problematic for reliable classification. Therefore the calculated results can be considered acceptable.

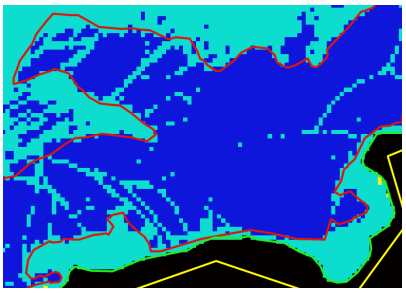
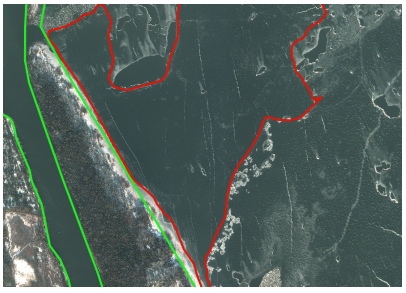
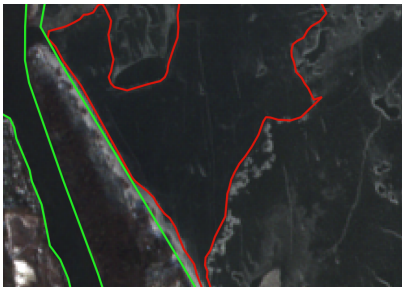
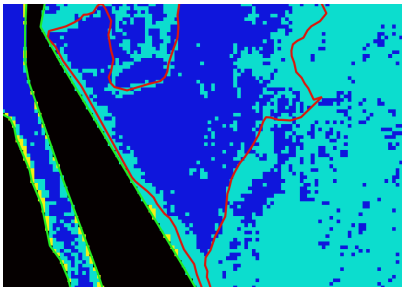
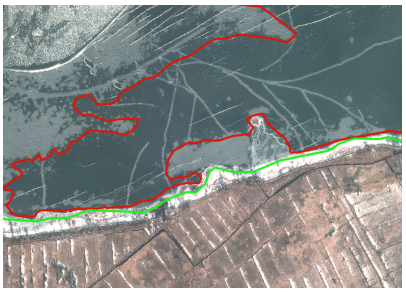
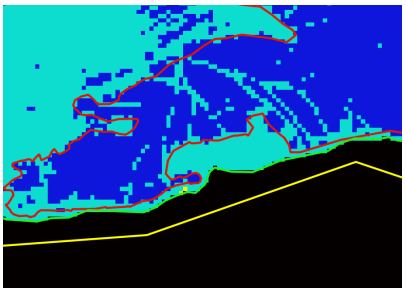
Pléiades 1A/1B	Sentinel-2	RLIE
		
		
		

Table 16. Particularly thin ice coverage underestimation. Comparison of Sentinel-2 RGB image, ICE classification (RLIE product) and Pléiades 1B RGB image, Lake Male Dabie, 27 January 2017, scale 1:10000. RLIE product: snow-covered or snow-free ice (cyan), open water (blue),

other features (yellow). Green lines delimit EU-Hydro extent. Red lines delimit manually drawn polygons for particularly thin ice.

MAJA Sentinel-2 L2A images

The confusion matrices between the nine MAJA S2 L2A images and their corresponding RLIE S2 products are given for lakes and rivers in Table 17 and Table 18 respectively. The derived quality indicators are presented in Table 19 for lakes and in Table 20 for rivers.

Results show that the overall accuracy of the ICE algorithm for RLIE S2 production is above the required level of 85%, with 98% for lakes and 99% for rivers. Kappa coefficients are at 83% and 98% for lakes and rivers respectively. Considering rivers and lakes water bodies separately, the 'snow-covered or snow-free ice' and 'other features' classes referring to lakes present the highest omission error. The underestimation of ice is connected with the limitation of particularly thin/dark ice detection. The omission rate for the 'other features' class results from the complexity of this class (vegetation and soil) and its challenging detection.

Users should note that this comparison is based on the same input data as used for the RLIE S2 production and its results should therefore be seen as slightly less reliable than the results obtained from the comparison with VHR images. However, this quality assessment shows that the threshold approach applied after the classification with the minimum distance classifier allows minimising misclassification errors due to algal bloom and turbidity and that the ICE algorithm can be considered a stable and reliable tool for the ice/snow detection over lakes as well as rivers.

MAJA Sentinel-2 L2A class	RLIE S2		
	open water	snow-covered or snow-free ice	other features
Open water	2109459 (91%)	44 (<1%)	76 (<1%)
Snow-covered or snow-free ice	54471 (2%)	147212 (6%)	84 (<1%)
Other features	2787 (<1%)	292 (<1%)	2990 (<1%)

Table 17. Confusion matrix between RLIE S2 product and MAJA Sentinel-2 L2A images referring to lakes

MAJA Sentinel-2 L2A class	RLIE S2		
	open water	snow-covered or snow-free ice	other features
Open water	49146 (84%)	1 (<1%)	50 (<1%)
Snow-covered or snow-free ice	2 (<1%)	7378 (17%)	24 (<1%)

Other features	251 (<1%)	74 (<1%)	1558 (3%)

Table 18. Confusion matrix between RLIE S2 product and MAJA Sentinel-2 L2A images referring to rivers

Quality indicator	RLIE S2 class		
	open water	snow-covered or snow-free ice	other features
Overall accuracy [%]	**97.5**		
Kappa coefficient [%]	**82.6**		
Omission error [%]	0.0	27.0	50.7
Commision error [%]	2.6	0.2	5.1
Recall [%]	100.0	73.0	49.3
Precision [%]	97.4	99.8	94.9
F1 score [%]	98.7	84.3	64.9

Table 19. Results of the validation of RLIE S2 products with MAJA Sentinel-2 L2A images referring to lakes

Quality indicator	RLIE S2 class		
	open water	snow-covered or snow-free ice	other features
Overall accuracy [%]	**99.3**		
Kappa coefficient [%]	**97.5**		
Omission error [%]	0.1	0.4	17.3
Commission error [%]	0.5	1.0	4.5
Recall [%]	99.9	99.6	82.7
Precision [%]	99.5	99.0	95.5

F1 score [%]	99.7	99.3	88.6

Table 20. Results of the validation of RLIE S2 products with MAJA Sentinel-2 L2A images referring to rivers

4. Conclusions

Algorithms generating RLIE S1 and RLIE S2 products have been assessed against in situ data and remote-sensing observations. Note that the independent observations (in situ or optical data) used in this report were acquired on the same date as the Sentinel source image of the RLIE product - or maximum 1 day apart for some observations. However in the case of ice phenomenon on rivers and lakes, the date and hour of observation has a major impact on the classification. For instance, river flows can carry the ice downstream or the ice structure can change within the day or the hour. In the frame of this assessment, it is assumed that the ice conditions remained nearly unchanged between Sentinel-1/Sentinel-2 and reference data acquisition points.

In view of this limitation related to the dynamic nature of the ice cover on the water surfaces, find the evaluation results summarised below.

4.1. ICE algorithm for RLIE S1 production

RLIE S1 products have been validated against a set of in situ observations (from the Czech Republic and Hungary) and manually classified Sentinel-2 L1C images.

The main conclusions of the RLIE S1 product validation are the following:

- The validation of the ICE algorithm for RLIE S1 production with in-situ data shows that the overall classification accuracy is at 76.7%. However, users should note that the accuracy for the class "ice" is only 25%. The detailed analysis of those results is provided in section 3.1.1.
- The validation of the ICE algorithm for RLIE S1 production with manually classified Sentinel-2 data shows that the overall classification accuracy is at 74.9%. However, users should note that more than half pixels identified as open water are detected as ice pixels by the RLIE S1 product (43% precision for water pixels). The detailed analysis of those results is provided in section 3.2.1.
- A comparison of RLIE S1 products with in situ observations reveals high omission and commission errors of the 'snow-covered or snow-free ice' class. Several causes were identified for the low classification accuracy: unreliability of in situ data (measurements often conducted near dams, measurements correspond to one particular point), limitations of the SAR sensors, lack of the 'other features' class in RLIE S1 product, changing ice conditions between time of the in situ observations and Sentinel-1 data acquisitions etc.
- The main limitation of the ICE algorithm for RLIE S1 production is the omission of particular ice forms (perfectly plain ice cover, snow-covered ice). Perfectly plain ice and ice covered with wet-snow are classified as 'open water' due to the similar surface scattering and physical properties to open water. For more information please refer to [AD4].
- Users should take into account that **the present RLIE S1 product assessment results are not applicable for lakes.** Only information located on rivers were considered and the classification by the ICE algorithm from Sentinel-1 observations has not been assessed on lakes. Ice detection on lakes from SAR imagery is indicated as the main limitation in the technical document [AD4] (section 2.1.2.8. Limitations), mostly because ice formed on lakes' surface is generally smooth. Further development of the

algorithm is needed in order to provide reliable ice information for lakes and other standing water bodies. Meanwhile users can rely on the QC-FLAGS layer of the RLIE S1 product to identify standing waters and discard them from their studies if necessary.

4.2. ICE algorithm for RLIE S2 production

The comparison and validation of the ICE algorithm for RLIE S2 production have been done against in situ measurements as well as six VHR optical remote-sensing images and nine manually interpreted MAJA Sentinel-2 L2A images.

The main conclusions of the RLIE S2 product validation are the following:

- The obtained validation results based on the comparison of RLIE S2 products with in situ measurements meet the requirement of the expected 85% accuracy level. As the in situ data are represented by single points (while single information of the RLIE S2 product refers to the minimum area of 20 m x 20 m) and are often conducted near dams or other hydraulic constructions, the provided information from in situ measurements and RLIE S2 products are compared for the established area. This limitation linked to the method can explain a decrease in the accuracy of the ICE algorithm.
- According to the quality assessment against VHR images, the ICE algorithm provides classification results over the expected 85% accuracy level. The ICE algorithm accurately detects ice coverage under various conditions with a slight tendency to underestimate the ice cover when water bodies are partially ice-covered by extremely thin ice.
- Validation against nine MAJA Sentinel-2 L2A images shows that the ICE algorithm provides stable results for rivers as well as lakes. Computed quality indicators are in both cases at a good level, above the required level of 85%.
- The results obtained from the quality assessment based on six VHR images and nine MAJA Sentinel-2 images as well as in situ measurements proved that the ICE algorithm is a reliable tool for ice coverage detection over inland waters.

5. Annex A: In situ data for RLIE validation

This annex presents the characteristics of the in situ datasets used for the RLIE validation. As in situ measurements vary by country, their structure and characteristics are described separately for each database.

5.1. In situ data in Czech Republic

In the Czech Republic - Povodí Vltavy - a state enterprise, is an administrator of water works in the Vltava river basin. They collect ice thickness data near dams on a daily basis during each winter season (mainly November - March). The ice cover type information is not included.

Data is provided in XLSX format and contains the station name, ice thickness value as well as the date of the observations. While comparing the in situ information with RLIE products, the ice thickness value above 0 is considered as ice cover on in situ measurements.

Data from 26 measuring ground station points was considered for RLIE S2 validation and 11 measuring ground stations for RLIE S1 validation - their spatial distribution is presented in Figure 12 and Figure 13 respectively. As the collection of in situ measurements is performed daily, only corresponding RLIE S1 and RLIE S2 products (acquired on the same date as in situ measurements) are taken into account during the scientific validation.

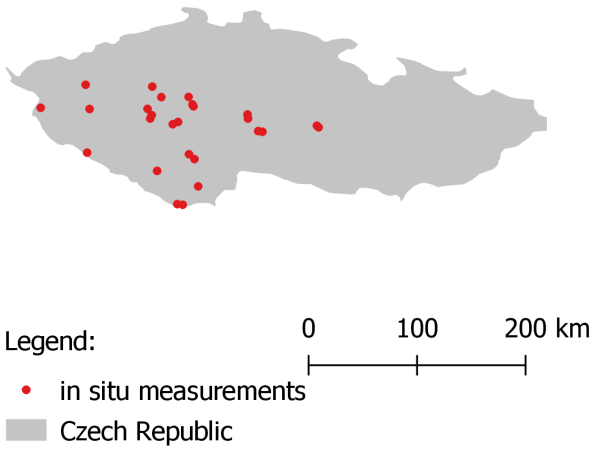


Figure 12. Spatial distribution of in situ measurements in Czech Republic used for the validation of the ICE algorithm for RLIE S2 production

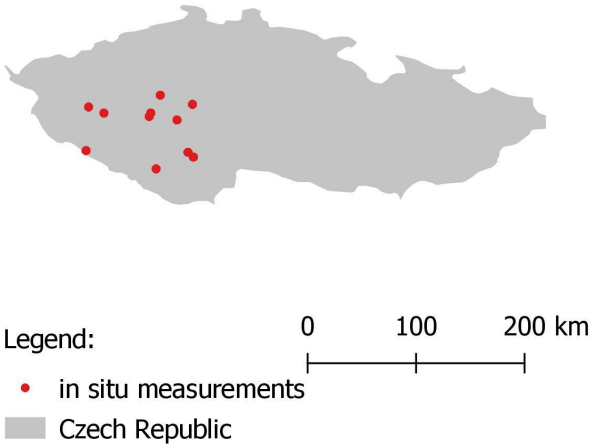


Figure 13. Spatial distribution of in situ measurements in Czech Republic used for the validation of the ICE algorithm for RLIE S1 production

5.2. In situ data in Finland

For more information about the collection of in situ data in Finland users should refer to [AUX1]. In situ data in Finland are collected by the Finnish Environmental Institute (SYKE). Ice thickness observations are taken manually three times per month (on the 10th, 20th and 30th day of the month) during the ice-covered season in winter (October-May, depending on the conditions that year). However, sometimes measurements are taken a few days before or after the aforementioned dates.

The provided data is tabulated; each record provides the following information:

- date of measurement
- total ice cover thickness (cm)
- congelation ice thickness (cm)
- white ice thickness (cm)
- water layer thickness between the ice cover (cm)
- white ice and water between the ice (cm)
- water surface in the drill hole (cm)
- snow depth (cm)
- white ice thickness based on white ice rods (frozen on the surface of the congelation ice layer)
- too-weak ice (unable to take measurements), 0 indicates that ice exists, but is too weak

In situ data from Finland is used only for RLIE S2 products validation as they refer mainly to lakes. For the evaluation of the RLIE S2 products, data are assumed as ice-covered when the total ice thickness is more than 0. Data is free of charge and can be downloaded from the website of the Finnish Environmental Institute after registration.

In Finland, in situ data for 27 measuring stations (see Figure 14) were used for the evaluation of the ICE performance. As the measurements are not performed on a daily basis, the corresponding RLIE S2 products with 1 day time lag were taken into account during the validation activities.

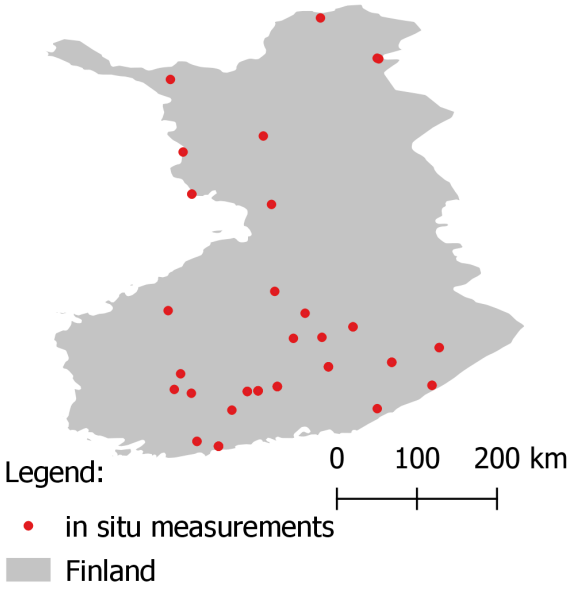


Figure 14. Spatial distribution of in situ measurements in Finland used for the validation of the ICE algorithm for RLIE S2 production

5.3. In situ data in Hungary

For more information about the collection of in situ data in Hungary users should refer to [AUX2].

Point-wise in situ data in Hungary is collected by the Hungarian Hydrological Data Centre. Measurements are made on a daily basis by visual observations near the stations during winter. The stations are located on water courses, mainly near dams. Data is provided in XLSX format (Microsoft Excel). All data is tabulated, and each record provides ice station name, internal station code, river name, coordinates in a defined projection, date of measurement, detected ice types and ice thickness. All ice types included during the measurements are described in another external file, also provided by the owner of the dataset. It is assumed that more than one ice type can occur. As the measurements include moving ice and other ice types unusable for the evaluation of RLIE products, all observations defined as 'ice drift', 'drifting ice cover' and 'moving ice cover' are excluded from the validation process.

All measurements are taken for 356 stations. Nevertheless, only 37 hydrological stations were considered for validation of the RLIE S2 product (see Figure 15) and 49 hydrological stations were considered for validation of the RLIE S1 product (see Figure 16) due to products limitations presented in section 2 and [AD3, AD4].

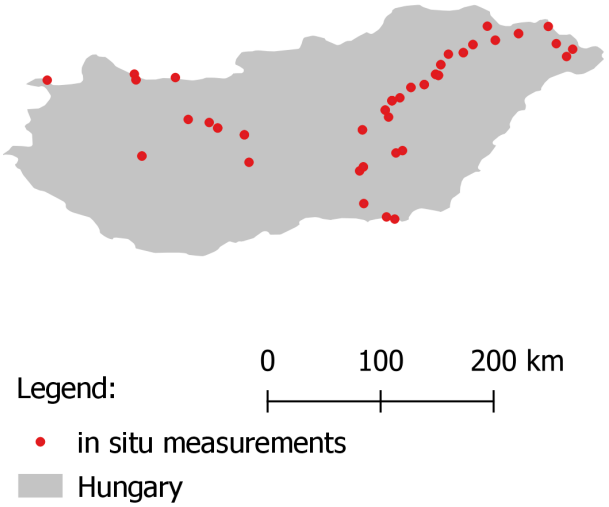


Figure 15. Spatial distribution of in situ measurements in Hungary used for the validation of the ICE algorithm for RLIE S2 production

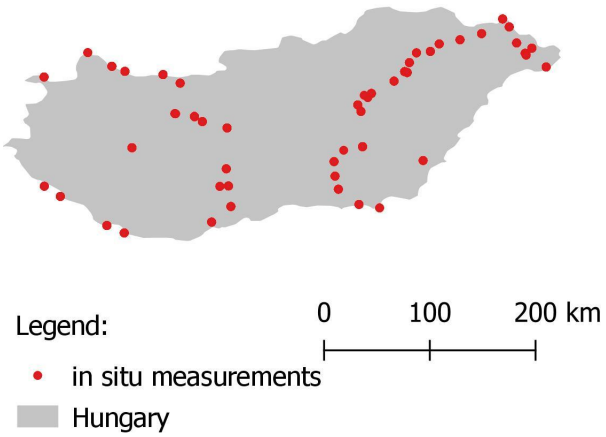


Figure 16. Spatial distribution of in situ measurements in Hungary used for the validation of the ICE algorithm for RLIE S1 production

5.4. In situ data in Serbia

The Republic Hydrometeorological Service of Serbia is in charge of observing ice cover at river stations in Serbia. The in situ measurements are collected on a daily basis during the winter

season (1st October - 31th March) at hydrological profiles, exclusively on rivers. There are no measuring stations on lakes.

In situ measurements are provided as an XLSX file with the information about the name and spatial coordinates of the measuring stations as well as observed ice phenomena: coastal ice, floating ice, ice pack. For each ice type the duration of each phenomenon (in days) is given as well as its start and end date. The ice thickness measurements are not performed.

In situ data from Serbia is used only for RLIE S2 products validation. During the validation, RLIE S2 products acquired during the occurrence of a particular ice type were used. Coastal ice and floating ice were excluded from the evaluation activities as their occurrence and extent are changeable in a short period of time and cannot be considered as reliable ice cover information for validation purposes. Comparing ice occurrence periods with the availability of cloudless RLIE S2 products (in the area of the measuring stations), for 7 hydrological stations the validation could be performed (see Figure 17).

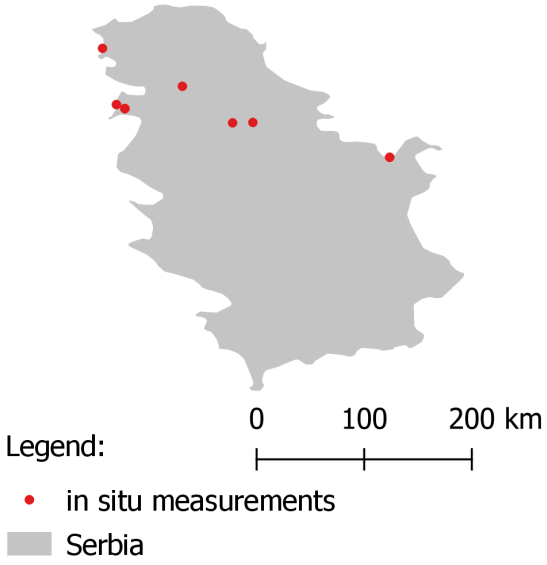


Figure 17. Spatial distribution of in situ measurements in Serbia used for the validation of the ICE algorithm for RLIE S2 production

6. Annex B: Optical remote-sensing images used for validation activities

This annex presents the dataset of remote-sensing images selected for the quality assessment of the ICE algorithm. The VHR optical remote-sensing dataset and the MAJA Sentinel-2 L2A images used for RLIE S2 assessment are detailed, as well as the set of Sentinel-1 L1C products used for the evaluation of RLIE S1 products. For all images, location, acquisition date and corresponding Sentinel-2 tile are specified.

No.	Site	Water body type	Reference product		Sentinel-2 RLIE product	
			Satellite sensor	Acquisition date	Tile	Acquisition date
1	Sweden (Orebro Lake)	lake	SPOT 6/7	31.03.2018	33VWF	31.03.2018
2	Norway (Nesvatn Lake)	lake	SPOT 6/7	31.03.2019	32VML	01.04.2019
3	Poland (Sniardwy Lake)	lake	SPOT 6/7	05.03.2018	34UEE	04.03.2018
4	Romania (Costesti River and Lake Costesti)	river and lake	SPOT 6/7	18.01.2019	35UMP, 35UNP, 35TNN	19.01.2019
5	Austria (Lake Waller, Irrsee Lake, Lake Obertrum)	lake	SPOT 6/7	27.02.2019	32UQU, 33UUP	28.02.2019
6	Poland (Male Dabie Lake)	river and lake	Pléiades 1A/1B	27.01.2017	33UVV	27.01.2017

Table 21. List of selected very-high-resolution images for RLIE S2 validation

No.	Site	MAJA Sentinel-2 L2A product	
		Tile	Acquisition date
1	Poland	34UCF	11.01.2017
2	Poland	33UYV	29.08.2019
3	Poland	33UXS	05.02.2019
4	Latvia	35VMC	16.02.2019
5	Germany / Austria	33UUP	28.02.2019
6	Romania	35TPL	26.08.2019
7	Hungary / Croatia	33TXM	31.08.2019
8	Sweden	34VCR	09.05.2020
9	Switzerland / France	32TLS	09.05.2020

Table 22. List of selected MAJA Sentinel-2 L2A images for RLIE S2 validation

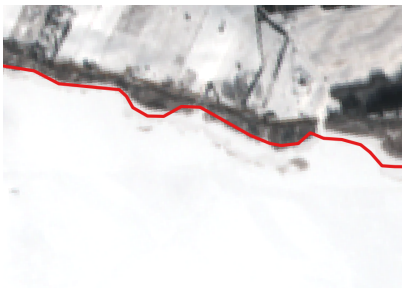
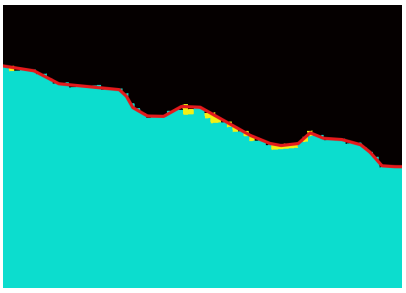
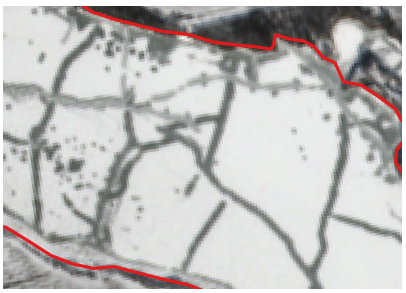
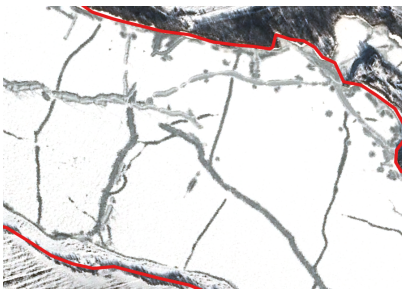
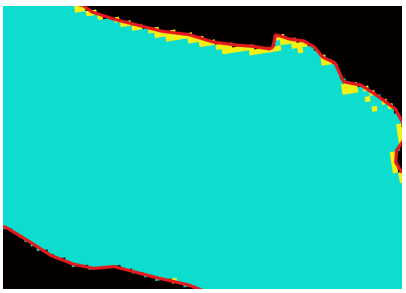
No.	Site	Sentinel-2 L1C product	
		Tile	Acquisition date
1	Lithuania	34UFG	17.01.2021
2	Lithuania	34UEG	17.01.2021

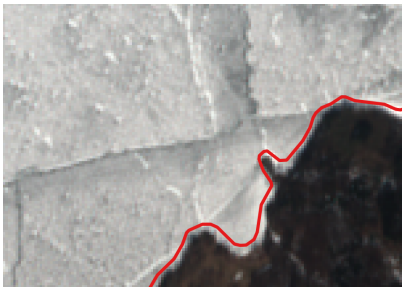
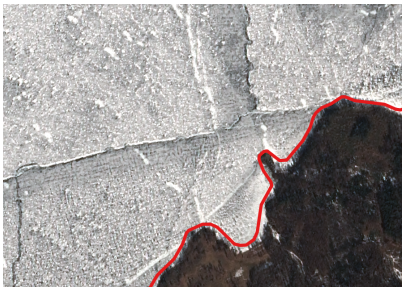
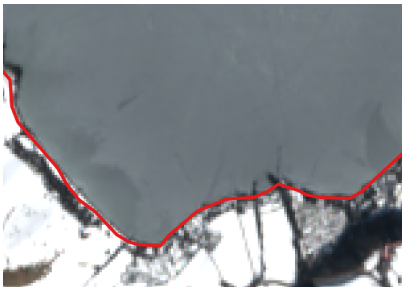
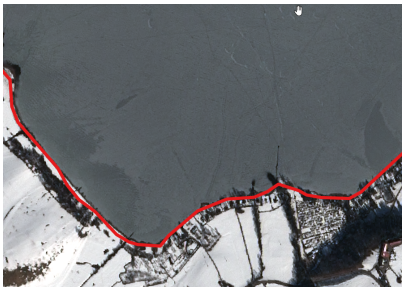
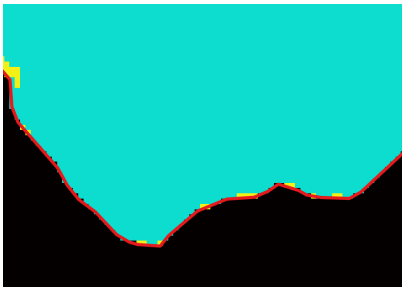
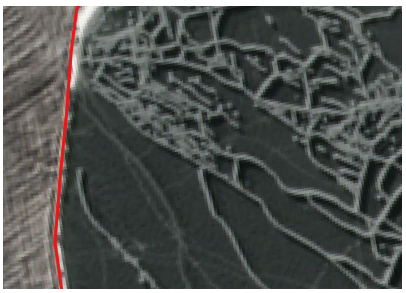
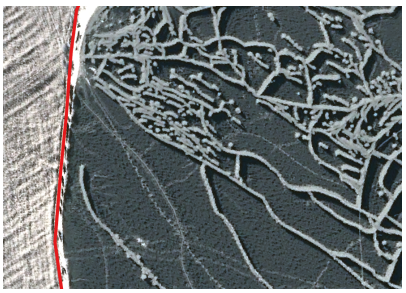
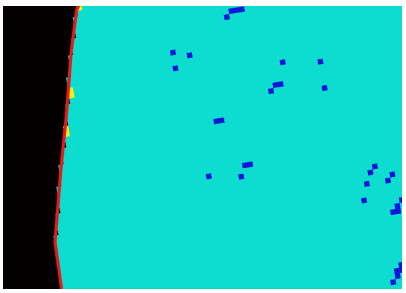
3	Lithuania	34UFG	14.02.2017
4	Norway	32VPP	04.02.2021
5	Norway	32VPN	04.02.2021
6	Norway	32VNP	04.02.2021
7	Poland	34UDD	08.01.2017
8	Serbia	34TDQ	15.01.2017

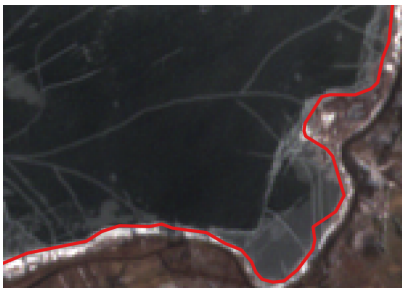
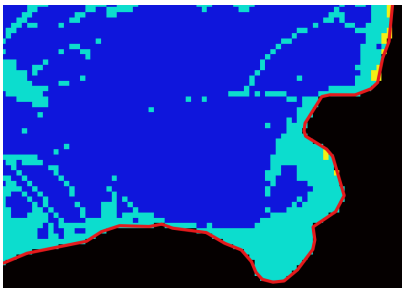
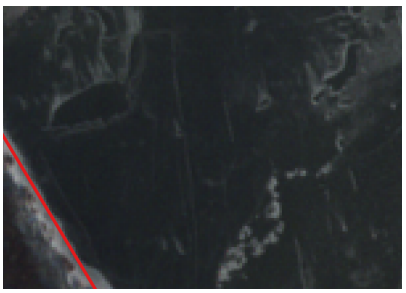
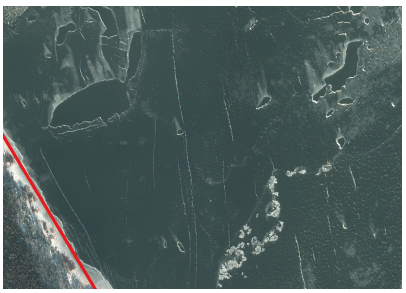
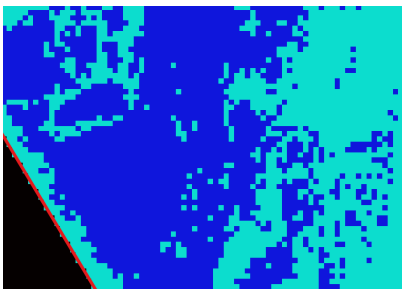
Table 23. List of selected Sentinel-2 L1C images for RLIE S1 validation

7. Annex C: Ice types review

This annex presents the different ice types taken into account while selecting the representative and reliable VHR images for the assessment activities.

Sentinel-2 RGB image	VHR RGB image	RLIE S2
Snow-covered ice		
		
		

Sentinel-2 RGB image	VHR RGB image	RLIE
Snow-free ice		
		
		
Broken ice		
		

Sentinel-2 RGB image	VHR RGB image	RLIE
Particular thin ice		
		
		

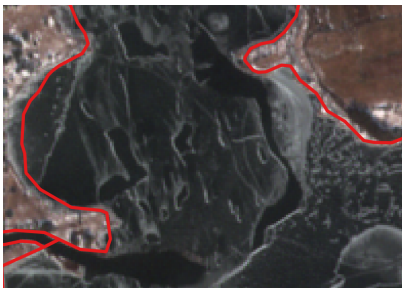
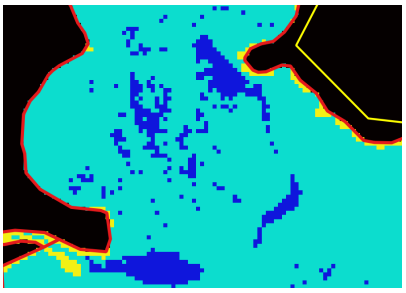
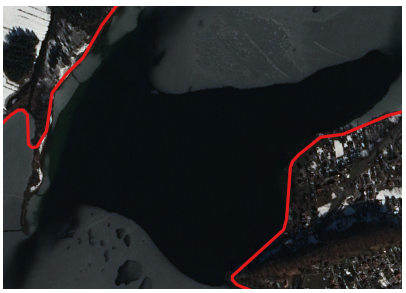
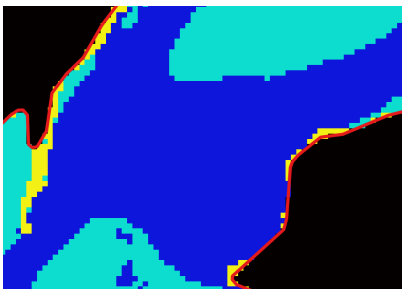
Sentinel-2 RGB image	VHR RGB image	RLIE
Partially ice cover		
		
		

Table 24. Examples of different ice types on VHR and Sentinel-2 images together with the RLIE product. Scale 1:8000. RLIE product: snow-covered or snow-free ice (cyan), open water (blue), other features (yellow). Red lines delimit the EU-Hydro extent.

8. Annex D: Calculation of evaluation metrics

This annex presents the formulas used to compute the quality indicators from the confusion matrices (see Figure 7 of the document):

- Overall accuracy defined by the proportion of pixels which were correctly classified to the overall number of pixels. Overall accuracy is calculated for the whole model:

$$Overall\ accuracy\ = \frac{n_{11}+n_{22}+n_{33}}{\Sigma n}$$

- Commission error calculated for each ICE class. An error of commission occurs when a pixel not assigned to a class is allocated to that class by the classification. Commission error for the 'snow-covered or snow-free ice' class is calculated according to the following formula:

$$Commission\ (ice)\ = \frac{n_{12}+n_{32}}{n_{12}+n_{22}+n_{32}}$$

- Omission error calculated for each ICE class. An error of omission occurs when a pixel assigned to a class is not allocated to that class by the classification. Omission error for the 'snow-covered or snow-free ice' class is calculated according to the following formula:

$$Omission\ (ice)\ = \frac{n_{21}+n_{23}}{n_{21}+n_{22}+n_{23}}$$

- Precision indicates the model's reliability level concerning the positive events regarding positive class assignment. It is calculated for each ICE class. Precision for the 'snow-covered or snow-free ice' class is calculated according to the following formula:

$$Precision\ (ice)\ = \frac{n_{22}}{n_{12}+n_{22}+n_{32}}$$

- Recall indicates the model's reliability level regarding positive event detection. It is calculated for each ICE class. Recall for the 'snow-covered or snow-free ice' class is calculated according to the following formula:

$$Recall\ (ice)\ = \frac{n_{22}}{n_{21}+n_{22}+n_{23}}$$

- F-measure / F1 score - the combination of recall and precision, calculated for each ICE class:

$$F\ -\ measure\ = 2\ *\ \frac{recall*precision}{recall+precision}$$

- Kappa coefficient [RD3] - indicates interrater reliability, calculated for the whole model:

$$\kappa = \frac{p_o - p_e}{1-p_e}$$

where:

$$p_e = \left(\frac{n_{11}+n_{12}+n_{13}}{\Sigma n} * \frac{n_{11}+n_{21}+n_{31}}{\Sigma n}\right)+\left(\frac{n_{21}+n_{22}+n_{23}}{\Sigma n} * \frac{n_{12}+n_{22}+n_{32}}{\Sigma n}\right)+\left(\frac{n_{31}+n_{32}+n_{33}}{\Sigma n} * \frac{n_{13}+n_{23}+n_{33}}{\Sigma n}\right)$$

$$p_o = accuracy$$